

Integrating and Evaluating Football Inspired Tactical Behaviours in RoboCup 2D Soccer Simulation

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Abstract

This project investigates whether football inspired tactical behaviours can be successfully integrated into the HELIOS Base architecture within the RoboCup 2D Soccer Simulator and improve overall team performance. A set of tactics were implemented including coordinated pressing, rest defence, a 4-2-3-1 formation with an asymmetric double pivot, adaptive attacking midfield play, sweeper keeper behaviour, set piece routines, and cutback based attacking patterns. The modified team, Ryan_FC, was evaluated against the original HELIOS Base team using quantitative and qualitative analysis across 330 simulated matches. The results highlighted significant improvements in win rate, possession, territorial control, and attacking output, demonstrating the effectiveness of the implemented tactical behaviours.

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1.0 Introduction and Problem Area

1.1 RoboCup Soccer Simulation

RoboCup is an international research initiative and competition that uses robotic and simulated football as a benchmark to advance research in artificial intelligence, robotics and multi-agent systems. “RoboCup’s original goal, as defined by its founding president Hiroaki Kitano, was to enable a team of fully autonomous humanoid robots to beat the best human soccer team in the world on a real, outdoor field by the year 2050.” [2].

Football was chosen as their research domain because it requires the integration of advanced technologies. These include teamwork, real time perception and decision, physical agents, and a high level of motion control [3]. Football also provides a complex decision making environment where the agents must react to rapidly changing situations, whilst only having partial information about the current state of the game [4]. In addition, as football is widely regarded as the most popular sport in the world, it provides a globally recognised and meaningful benchmark for technological advancement [3].

The RoboCup Soccer Simulation league “is currently split into two competitions called 2D Soccer Simulation League and 3D Soccer Simulation League.” This project focuses on the 2D league where two teams of 11 “agents” play a virtual game of soccer in a stadium. [5] Each agent independently controls a single player and they interact with the Soccer Server through a client-server architecture [6].

RoboCup’s Soccer Simulation environment (known as Soccer Server) has become an important platform for research in artificial intelligence, especially for studying multi-agent systems. The Soccer Server was created to act as a benchmark environment for evaluating multiagent systems and cooperative algorithms [4]. The platform enables researchers to produce large scale and reproducible experiments, whilst evaluating different strategies in parallel which allows for efficient testing [7]. Finally, the Soccer Server eliminates the need for physical robots which removes many constraints associated with physical robots, such as hardware development. This in turn allows for research to solely focus on the issues related to simulated football and artificial intelligence, such as decision making and cooperation [6].

1.2 The HELIOS RoboCup Team

HELIOS is a simulated soccer team which participates in RoboCup 2D competitions. The HELIOS team are extremely successful in the RoboCup 2D Simulation League. The team first competed in the RoboCup competition in 2000 and has since won 8 championships, consistently placing amongst the top four teams every year since 2005 [8], [9], [10].

HELIOS, like other RoboCup teams, consist of multiple autonomous agents where each agent controls a player in the simulation. However, HELIOS uses a layered decision making structure to determine how

each individual player behaves throughout the match. This structure is divided into three main parts, the agent class, role class and behaviour class. The agent class is responsible for determining the team's overall strategy and assigning a role to each player. Based on this role, the role class evaluates the current game situation and selects the best corresponding tactical behaviour. Finally, the behaviour class carries out the specific action needed such as moving, passing or shooting [11].

HELIOS Base was selected as the foundation for this project for various reasons. Firstly, it has been widely adopted by the RoboCup 2D Simulation League community and is continuously being used by many researchers [12]. A previous study has shown that HELIOS Base became the most widely used code base in the League, with 83.3 percent of teams that competed in 2013, using HELIOS Base as a framework for their development [11]. Several successful teams have built their teams on top of the HELIOS architecture, for example Gliders and Fractals have used the Agent2D framework and have then gone on to win the RoboCup championship in 2016 and 2019 respectively [12]. Also, HELIOS itself has continued to be a highly competitive team. They have won the last four RoboCup 2D championships in a row [10]. Finally, HELIOS Base has been selected as a starting point, due to its modular and role based architecture which allows for individual player behaviours to be modified [11]. This is particularly useful for implementing real world tactical behaviours where players roles can be adapted to reflect modern tactical responsibilities.

1.3 Problem Area

Although the RoboCup Soccer Simulation League has played an important role in advancing research in multi-agent artificial intelligence and cooperative systems [4], [13], many RoboCup teams rely on algorithmic decision making frameworks in which the agents evaluate possible actions based on the current state of the game before executing an action [11], [12]. While these methods do allow agents to operate effectively in real time, they don't always represent the structured tactics used by real world football teams.

RoboCup's long-term objective requires agents to display intelligence and coordination comparable with professional football players, as their ultimate goal is to beat the most recent FIFA World Cup winners by 2050 [14]. Achieving this objective will rely on advances in robotics and artificial intelligence, including decision making and teamwork. However, the tactical organisation shown in real world football may provide useful insights into how the coordination behaviour of agents can be achieved and improved.

In professional football, tactical systems organise player movement and positioning across the pitch to create scoring opportunities. Teams function as a coordinated system where their collective interactions and spatial organisation between each other influence the overall attacking and defensive performance [15], [16]. These strategies have evolved throughout the years by meticulous analysis and

application by coaches and analysts. Previous research has suggested that incorporating human inspired tactics into robotic soccer systems may provide valuable improvements. Research further highlighted how RoboCup teams rarely employ predefined strategies to coordinate movement and passing actions, despite how commonly it is used in human team sports. This suggests that there is an opportunity to take this knowledge used by football teams and implement it within the RoboCup 2D Simulation League [17].

However, implementing real world tactics into RoboCup agents is a significant challenge. The tactics developed must be converted into algorithms that work in a fast-paced changing environment, where each agent makes its own decision based on the available information it has at that current point in the match. A previous research experiment has shown that adding real world tactics can reduce the team's overall performance. The updated HELIOS Base team only achieved a 10% win rate against the original HELIOS Base Team [18], demonstrating how challenging it is to integrate human football tactics into RoboCup.

This project aims to not simply add real world tactics to a RoboCup team, but instead will assess whether these tactics can be implemented in a way that improves the overall performance of the team, without weakening the original strengths of the baseline system. To achieve this, the project will first analyse how the original HELIOS Base team behaves, then integrate a new set of tactics to create improvement which will be tested to determine whether match performance improved.

To address this challenge, the project focuses on implementing a range of tactical behaviours inspired by modern professional football. These include coordinated pressing strategies, structured formations (such as a 4-2-3-1 system), and role specific behaviours including adaptive attacking midfield play and advanced goalkeeper decision making.

In addition, the project explores situational tactical concepts such as rest defence, set piece exploitation, and cutback based attacking patterns, alongside the integration of performance analysis techniques to evaluate the effectiveness of these behaviours.

1.4 Analysis of Previous Approaches

Review of previous implementation of modern football tactics into the HELIOS Base team, illustrates a key limitation. The system was using predominantly reactive decision making behaviours rather than maintaining a consistent tactical identity. The system adjusted its behaviour based on match state variables such as the scoreline. An example of this was that pressing intensity increased mainly when the team was losing, whereas they adopted more conservative behaviours when the match was level or the team were ahead [18]. This may have resulted in the team allowing the opponents to dictate the tempo of the game until a negative match state forced a response.

Another possible limitation was observed in the attacking phase through reliance on an expected goals (xG) model for shot selection. Although xG provides a useful estimate of shot quality, it appears to have encouraged more passive behaviour where agents delayed or avoided shooting opportunities to instead retain possession. This leads to a situation where the agents are overly cautious and wait for optimal conditions before shooting which can reduce the number of scoring opportunities.

However, in real football important actions do not always arise from analytically optimal situations. Strict adherence to xG based decision rules may undervalue rare but potentially decisive opportunities, particularly in high pressure contexts where decision windows are brief. When designed computationally the system could become too cautious and miss valuable chances. The issue is not how shots are executed physically, but how the decision to shoot is made in the first place.

This analysis has led to a revised design of the project that is more proactive, rather than relying on reactive triggers, making decisions based on the situation instead of strictly adhering to statistics.

1.5 Project Aim and Objectives

This project investigates whether tactics inspired by professional football can be integrated into the HELIOS Base architecture within the RoboCup 2D Soccer Simulation. The aim of the project is to implement carefully selected tactical behaviours based on real world examples and then evaluate their impact on the overall performance of the simulated team.

To achieve this aim, the following objectives have been created:

1. Analyse the RoboCup 2D Soccer Simulation environment and the HELIOS Base architecture to understand how agents make their decisions and how individual player behaviours can be modified within this existing framework.
2. Investigate tactics used by top performing professional football teams, particularly in relation to player positioning, coordinated movement and team organisation, in order to identify potential strategies that could improve the performance of the HELIOS Base 2D RoboCup team.
3. Design algorithms based on objective 2 that convert the selected real world football tactics into behaviours that RoboCup agents can execute.
4. Implement the chosen tactical behaviours identified into the HELIOS Base codebase by modifying the relevant classes that control the agent's decision making.
5. Conduct controlled simulation experiments in the RoboCup Soccer Server to compare the performance of the modified team against the original HELIOS Base team under consistent conditions.
6. Evaluate the impact of the implemented tactics using measurable metrics such as match outcomes and goals scored versus conceded and interpret whether these changes have improved the overall team performance.

Together, these objectives aim to determine whether tactical concepts inspired by real world football can be successfully implemented within the RoboCup 2D Soccer Simulator and if they can improve the performance of the HELIOS Base team.

2.0 System Requirements and Specification

2.1 Solution Description

This project extends the existing HELIOS Base RoboCup 2D simulation team by integrating tactics inspired by modern professional football. The system adds structured decision making frameworks across all phases of the match with the aim of improving coordination, positioning and overall team performance.

The final system focuses on implementing both team level tactics such as formations, rest defence and agent level behaviours including goalkeeper positioning and adaptive attacking roles. The system also includes statistical analysis to evaluate performance through measurable metrics and visualisations. The system is designed to operate within the constraints of the RoboCup 2D Soccer Simulation environment and remain compatible with the HELIOS Base architecture.

2.2 Assumptions and Constraints

Assumptions:

- Each agent operates autonomously with access to only partial and local information provided by the server.
- The opponent team (HELIOS Base) is assumed to use predefined decision making logic and can be used as a consistent benchmark for evaluation.
- Although the RoboCup simulation includes stochastic elements such as noise in perception and actions, it is assumed to be sufficiently consistent for evaluation.
- Tactical behaviours are modelled using rule based or heuristic decision making rather than complex learning methods.
- The 2D simulation environment is simplified but provides a realistic model of football for evaluating tactical behaviours.

Constraints:

- All decision making from the agents must occur within the real time constraints of the RoboCup simulation cycle.
- Agents must operate using decentralised decision making which makes coordination between players from local behaviours rather than a central control mechanism.

- Agents do not have access to global game state information and must make decisions based on local and uncertain data.
- Modifications must be implemented within the existing HELIOS Base codebase, limiting architectural changes.
- Agents have stamina and physical limitations defined by the simulator, which eliminates the use of continuous high intensity actions.
- The system must remain compatible with the Robocup Soccer Server (rcssserver) and the RoboCup Monitors (rcssmonitor and SoccerWindow2).
- The simulation doesn't fully replicate real world football which could affect the effectiveness of certain tactical behaviours.

2.3 Functional Requirements

The functional requirements describe how the system should behave. Each of the requirements are designed to be testable within the simulation and can be evaluated to analyse their effectiveness.

Team Tactical Behaviour

- The system shall implement a coordinated pressing strategy in advanced areas, where pressing is activated when the opponent is likely in possession and enough agents are within a support range.
- The system shall coordinate multiple agents to apply collective pressure which includes cutting out passing lanes and supporting the primary presser.
- The system shall maintain rest defence positioning to ensure that the deeper players remain structured and don't end up out of position.
- The system shall adjust the teams shape based on game context and the ball location.

Justification: These behaviours aim to improve possession recovery in advanced areas to create more pressure leading to an increase in shots/goals whilst also reducing the vulnerability to counter attacks.

Team Formation and Organisation

- The system shall use a 4-2-3-1 inspired formation for team structure, with an asymmetrical midfield configuration. One defensive midfielder operates in a deeper more defensive role, whereas the second midfielder has a more advanced position in order to support attacks.
- The system shall assign each agent a role with each agent operating based on the role and positioning from the formation.
- The system shall stay compact defensively, however then create width when attacking using the wide forwards and the other advanced players, with players positions adjusting based on the ball's location.

Justification: The 4-2-3-1 formation is used to improve control in the midfield whilst providing defensive balance and it also allows for a wide range of attacking options.

Player Role Behaviour

- The system shall use HELIOS Base's role-oriented agent structure with modifications applied to selected roles.
- The system shall implement improved decision making for the central attacking midfielder based on local pressure, passing lane quality, teammates positioning and overall attacking context to provide the most intelligent action such as a through ball, controlled dribbling or a wide pass.

Justification: The customised roles are used to improve agent intelligence to create better decision making, leading to more chances and better quality chances in the match.

Goalkeeper Behaviour

- The system shall implement a predictive goalkeeper positioning based on ball location and the shooting angle.
- The goalkeeper shall perform sweeper keeper actions to advance from the goal line to clear danger behind the defence when it's safe to do so.
- The goalkeeper shall be a passing outlet during build up play when the team has possession and there is no immediate defensive danger.

Justification: These behaviours aim to reduce goals conceded, improve defensive coverage in dangerous situations and to improve the stability of the team during buildup phases.

Attacking Patterns

- The system shall implement cutback based attacking scenarios.
- The system shall coordinate attacking movement to create high quality shooting opportunities.

Justification: Cutbacks have been chosen as they disrupt defensive organisation by forcing defenders towards their own goal before reversing the play which creates time and space for attackers in a dangerous central area to get a high quality shot off.

Performance Evaluation

- The system shall record all match outcomes, including wins, draws and losses to allow for overall performance comparison.
- The system shall track the number of goals scored and the number of goals conceded from each match. Goal difference shall be calculated from these values to compare the attacking effectiveness and the defensive stability.

- The system shall collect tactical performance metrics to evaluate the effectiveness of the chosen implemented behaviours including the number of saves, overall possession and number of shots.

Justification: The metrics will provide clear evidence to be able to evaluate the overall team performance, in terms of both match outcomes and the effectiveness of each individual implemented tactic.

2.4 Non-Functional Requirements

Non-functional requirements focus on the systems performance rather than what the system actually does. These requirements ensure the system operates reliably and efficiently, and that it can be effectively integrated, tested, and evaluated within the RoboCup simulation environment.

The following non-functional requirements are:

- Performance: The system shall operate within the real time constraints of the RoboCup simulation cycle.
- Reliability: The system shall operate consistently across all simulation runs without unintended tactical behaviours.
- Stability: The system shall run all simulations without errors or crashes.
- Compatibility: The system shall ensure full compatibility with the HELIOS Base architecture.
- Maintainability: The system shall support the addition, modification or removal of tactics without breaking existing system functionality.
- Consistency: The system shall produce consistent results over multiple simulation runs, even with the server's randomness.

These nonfunctional requirements ensure the system is not only functionally effective but also robust to ensure its suitability for evaluation.

2.5 System Interfaces

- The system shall interact with the RoboCup Soccer Server (rcssserver) to run simulation matches.
- The system shall use visualisation tools including rcssmonitor and SoccerWindow2, to observe and analyse matches and agent behaviour.
- The system shall be integrated with the HELIOS Base codebase to improve agent decision making and overall team coordination.
- The system shall generate and output match data and performance metrics to support analysis of the games and the evaluation of the team's performance.

2.6 User Characteristics

- Users of this system are expected to have a basic understanding of the RoboCup simulation environment and familiarity with fundamental programming concepts especially in C++ as the system is built using the HELIOS Base codebase.
- Typical users may include students, developers and researchers who have an interest in multi-agent systems or robotic football simulations, particularly those interested in analysis and implementing football tactics.
- Users should be able to run the simulation, edit tactics and analyse the data outputted by the system.

2.7 Scope and Limitations

The system is not intended to fully replicate the complexity of real world football, instead it focuses on evaluating the chosen selected tactics and their effectiveness within the constraints of the RoboCup 2D simulation environment.

The RoboCup environment operates in a simplified two-dimensional space. It includes randomness such as partial observability, which could cause variability in match outcomes.

Agents within the simulation have physical and behavioural limits. Their physical attributes such as speed and stamina are fixed by the environment and can't be improved through training. Accompanying this, the agent's behaviour is determined by the programmed logic which means agents can't learn or adapt based on experience but instead rely solely on the rules implemented by the system.

This means that the system's performance depends on the quality of the implemented tactics and the outcomes may vary, due to the randomness of the simulation and interactions between agents during matches. Therefore, the results should be considered within the context of the simulation and not as a direct reflection of real world football performance.

2.8 Use Case Descriptions

The following use cases were derived from the functional requirements to show how the behaviours are executed within the simulation.

UC1: Coordinated Pressing

Preconditions: Simulated match is running and the opponent is likely in possession of the ball.

Objective: To apply coordinated team pressing in order to disrupt opponent possession.

Actors: Ryan_FC's outfield agents.

Trigger: The ball enters a predefined pressing zone and enough agents are within range.

Main Flow:

1. The system evaluates whether pressing conditions are satisfied based on ball location and agent proximity.
2. If conditions are met, a coordinated press is initiated.
3. The nearest agent applies direct pressure to the ball carrier.
4. Supporting agents reposition to block passing lanes and maintain compactness.
5. Defensive structure is preserved by limiting excessive forward movement.

Postconditions:

Pressure is applied to the opponent while maintaining defensive balance.

UC2: Structured Set Piece Execution

Preconditions: A set piece situation occurs (e.g. free kick, corner, or restart).

Objective: Execute set plays that balance attacking opportunities with possession retention.

Actors: Ryan_FC's agents.

Trigger: The simulation enters a set piece state.

Main Flow:

1. The system identifies the type and location of the set play.
2. A set of candidate actions (pass, cross, shot, recycle) is generated.
3. Potential receivers are evaluated based on positioning and pressure.
4. The optimal action is selected based on safety and attacking potential.
5. If no strong option is available, possession is safely retained.

Postconditions: Play is restarted in a controlled and tactically appropriate manner.

UC3: Predictive Goalkeeper Positioning

Preconditions: Opponent is in an attacking position.

Objective: Improve goal coverage by predicting likely shot trajectories.

Actors: Ryan_FC's Goalkeeper agent.

Trigger: A dangerous attacking scenario is detected.

Main Flow:

1. The system evaluates threat level based on ball position and opponent proximity.
2. A likely shooting angle is estimated.
3. The goalkeeper adjusts position to maximise goal coverage.
4. Position is dynamically updated as the situation evolves.

Postconditions: The goalkeeper is optimally positioned prior to a shot.

UC4: Sweeper Keeper Support

Preconditions: The team has control of the possession.

Objective: To support buildup play using the goalkeeper.

Actor: Ryan_FC's goalkeeper agent

Trigger: Ball is in defensive or middle third with low risk.

Main Flow:

1. The system evaluates safety conditions.
2. The goalkeeper advances into a supporting position.
3. The goalkeeper becomes available as a passing option.
4. The goalkeeper retreats if danger increases.

Postconditions: Buildup play is supported without compromising defensive security.

UC5: Formation Based Team Structure

Preconditions: Simulation match is running

Objective: To maintain a structured formation that supports both attack and defence.

Trigger: Match is in play

Main Flow:

1. Formation positions are assigned based on configuration data.
2. Player positions adjust dynamically relative to ball location.

Postconditions: Team shape remains organised.

UC6: Asymmetric Double Pivot Midfield

Preconditions: Ryan_FC is in possession.

Objective: To provide midfield balance and progression.

Actors: Ryan_FC agents 6 and 7

Trigger: Match is in play

Main Flow:

1. One midfielder maintains a deeper position.
2. The second midfielder operates slightly higher.
3. Positional conditions are evaluated to confirm asymmetry.
4. Midfield structure supports both defence and attack.

Postconditions: A stable and progressive midfield structure is maintained.

UC7: Rest Defence Maintenance

Preconditions: The team is pressing or attacking.

Objective: To maintain defensive compactness during attacking phases.

Actors: Ryan_FC's outfield agents

Trigger: The team is in an attacking or pressing phase.

Main Flow:

1. Non-pressing agents maintain deeper positions.
2. Players shift toward the ball while preserving structure.
3. Central defensive zones are prioritised.

Postconditions: The team remains protected against counterattacks.

UC8: Adaptive Attacking Midfielder Decision Making

Preconditions: The central attacking midfielder is in possession or near the ball.

Objective: To select context aware attacking actions.

Actors: Ryan_FC's agent 8

Triggers: The player is in possession or involved in attacking play.

Main Flow:

1. The system evaluates pressure, space and passing options.
2. Candidate actions are generated (pass, dribble, shoot).
3. Each action is scored heuristically.
4. The highest value action is selected.
5. Fallback behaviour is applied if needed.

Postconditions: Attacking decisions are context sensitive and efficient.

UC9: Wide Attacking and Cutback Play

Preconditions: The ball is in wide attacking areas.

Objective: To create high quality chances using cutbacks and wide play.

Actors: Ryan_FC agents 9 and 10

Trigger: The ball enters wide attacking areas.

Main Flow:

1. Target zones are identified (penalty spot, near/far post).
2. Passing lanes and pressure are evaluated.
3. The optimal attacking action is selected.
4. Supporting players position accordingly.

Postconditions:

Attacks generate higher quality scoring opportunities.

UC10: Performance Data Collection and Analysis

Preconditions: Matches have been simulated.

Objective: To evaluate system performance using statistical metrics.

Actors: Analysts performing simulation analysis

Trigger: The analysis process is initiated when the analyst executes the data processing scripts following the generation of match log (rcg) files.

Main Flow:

1. Match data is collected from simulation logs.
2. Metrics are extracted and calculated.
3. Results are aggregated across matches.
4. Outputs are generated for evaluation.

Postconditions: Performance data is available for analysis and comparison.

3.0 Design

The design stage uses the system requirements defined in section 2 and converts them into a structured and implementable solution. This is especially important in RoboCup because it is a multi-agent environment, where team coordination is based on individual agent behaviours.

This system is built upon the HELIOS Base architecture, where selected behaviours and decision making logic have adapted to introduce real world football tactics. Rather than redefining the entire system, this project focuses on adding tactical intelligence within the existing stable framework.

The following subsections describe the system architecture, key behavioural components and the rationale behind the design decisions.

3.1 Existing HELIOS architecture and design approach

The HELIOS Base team follows a layered, role based agent architecture where each player operates as an autonomous agent within the RoboCup 2D simulation environment. Each agent is assigned a role that controls its decision making and movement, whereas team level organisation is controlled through a strategy system and formation data.

The system can be divided into four parts which interact with one other. Firstly, the simulation environment provides real time state and enforces the rules of play. Secondly, the strategy and formation layer determines the overall tactical structure of the team. Thirdly, the role layer defines what each individual agent is responsible for and finally the behaviour layer executes specific actions based on the current game state including moving, passing and shooting. This structure allows effective team coordination together through local decision making.

The system operates in a continuous loop, where agents perceive the environment, evaluate the current game state, and select actions at each simulation cycle. This is essential due to the simulation

running real time, where multiple agents must make decisions at the same time without a central controller.

Instead of redesigning the whole system, this project aims to improve selected parts of the HELIOS Base team by adding football tactics whilst maintaining the stability and reliability of the original framework. A complete redesign would significantly increase the development complexity and risk breaking an already functional team. By extending the current system, it ensures that the project remains compatible with the baseline, while allowing new tactics to be added in a controlled and testable manner. The architecture of the HELIOS Base system, including the integration of the proposed tactical extensions, is illustrated in Figure 3.1.

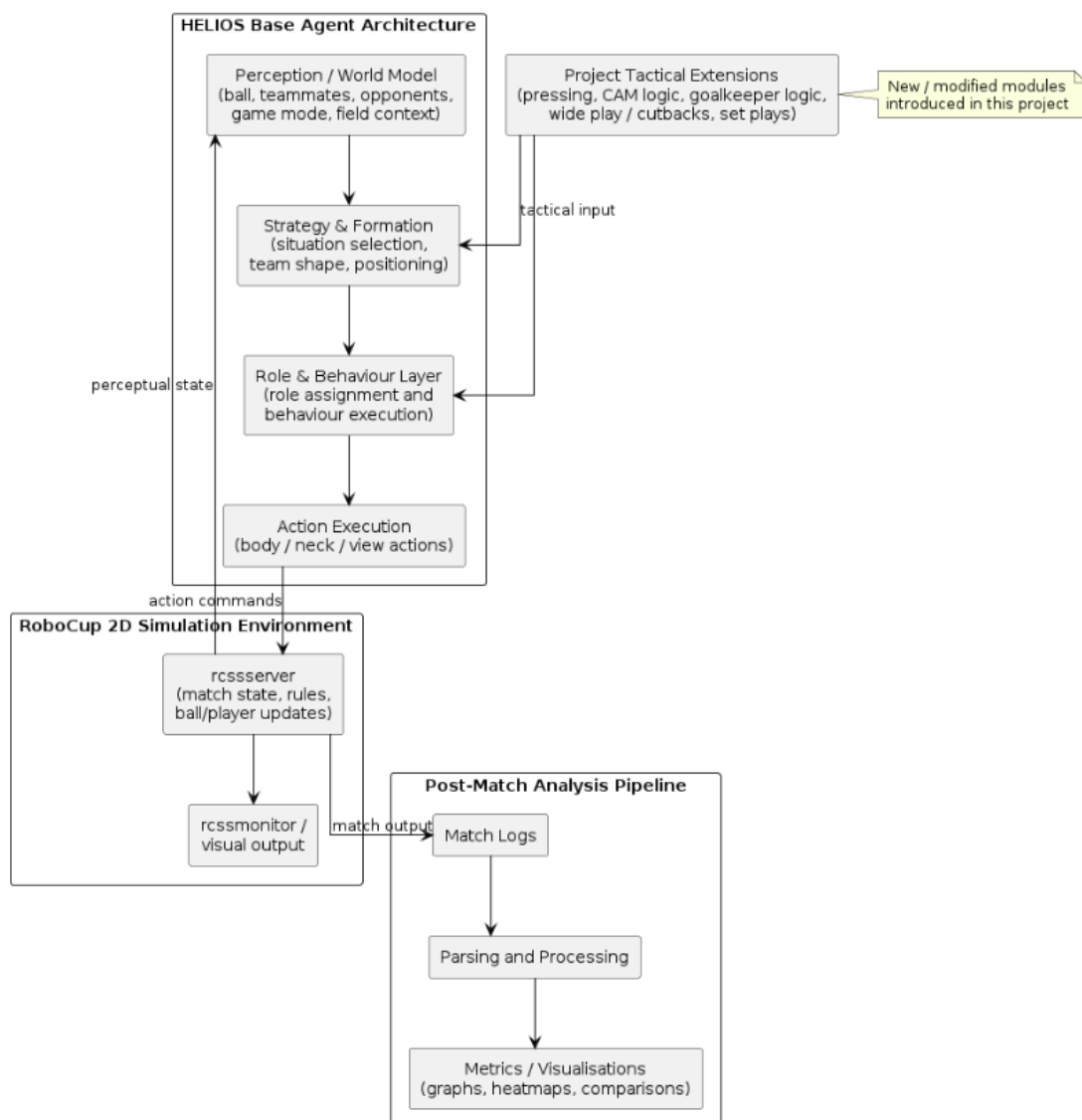


Figure 3.0.1 HELIOS Base agent architecture showing perception, strategy, role, and behaviour layers, alongside tactical extensions and post-match analysis pipeline

3.2 Design Method: Use Case Driven Tactical Extensions

The design of the modified system was guided by a set of use cases. Each use case represented a football situation that the team should execute in a more realistic or effective manner than the original HELIOS Base team. Instead of designing the team as one large system, the project was broken down into smaller manageable behaviours such as pressing, goalkeeper support, cutback creation and set pieces.

This approach was useful because it allowed the system to be developed around real match situations. It also created a clear link between the requirements and the implemented behaviours ensuring that each proposed design component directly contributes to the overall project objectives. The project's main aim of the design is not to create a new system but instead integrate multiple modern football tactics into the existing framework. The mapping between the defined use cases and their corresponding system modules is shown in Table 3.1.

Table 3.1: Mapping of defined use cases to associated system modules

Use Case	Associated Files
UC1 Coordinated Pressing	bhv_basic_move.cpp, sample_player.cpp
UC2 Structured Set-Piece Execution	bhv_set_play_free_kick.cpp, bhv_set_play_kick_in.cpp, bhv_set_play_kick_off.cpp
UC3 Predictive Goalkeeper Positioning	bhv_goalie_basic_move.cpp
UC4 Sweeper-Keeper Support	bhv_goalie_basic_move.cpp
UC5 Formation-Based Team Structure	defense-formation.conf, normal-formation.conf, offense-formation.conf
UC6 Asymmetric Double Pivot Midfield	defense-formation.conf, normal-formation.conf, offense-formation.conf
UC7 Rest Defence Maintenance	bhv_basic_move.cpp, sample_player.cpp
UC8 Adaptive Attacking Midfielder Decision-Making	role_offensive_half.cpp
UC9 Wide Attacking and Cutback Play	role_side_forward.cpp
UC10 Performance Data Collection and Analysis	log files (rcssserver output), parsing scripts, analysis / visualisation scripts

3.3 Formation and midfield structure design (UC5, UC6)

The first design objective was to introduce a tactical shape inspired by real world football. The system adopts a 4-2-3-1 formation which is widely used due to its balanced approach and effectiveness in midfield control [19]. This formation is widely used in modern football with 13 Premier League teams adopting it as their primary formation during the 2024/25 season [20]. The use of a double pivot provides a solid midfield base and supports the link between defensive and attacking phases of play [21].

In this project an asymmetrical double pivot was selected over a flat midfield pairing, meaning one midfielder remains deeper to act as a stabilising defensive player, while the other is positioned slightly higher in order to support ball progression and connect the midfield to the attack.

One important design consideration was that formations in a simulated environment are dynamic and constantly change, due to agent movement which is influenced by many factors including ball position, opponent structure and local decision making behaviour. As a result, the formation may vary

throughout the duration of the match rather than maintaining a fixed structure because the design prioritises functional positioning and role effectiveness over strict adherence to fixed formation positions. The formation structure and its supporting system design flow are illustrated in Figure 3.2.



Figure 3.2: (a) 4-2-3-1 formation with asymmetric double pivot structure. (b) Formation processing pipeline from strategy component to behaviour layer.

3.4 Pressing and Rest Defence (UC1, UC7)

The defensive design of the system combines pressing with rest defence structure to balance aggressiveness with defensive stability.

The pressing system is designed to be a coordinated team behaviour rather than an individual player chasing the ball. The press is controlled by conditions including who is in possession, ball location and the number of players within supporting range. This ensures that the pressing is coordinated and only triggers strategically rather than continuously.

When the press has been activated the system assigns set roles to the agents. The closest player to the ball becomes the primary presser and nearby players block passing lanes, whilst the rest of the agents shift as a unit towards the ball to create compactness around the ball. This allows the team apply pressure while remaining defensively solid.

Rest defence complements pressing, as when the pressing is active defensive agents will stay back, remain in shape and not get dragged out to chase after the ball. Instead, agents will position themselves in areas to block the most likely movement of the opponents. This ensures that the team cover critical areas and prevent vulnerability to counter attacks.

This design reflects the nature of the system where multiple agents are executing their own specific pressing behaviour to coordinate together as one unit. To ensure the stability of the team, constraints

were added to help maintain the defensive structure and to prevent the team from becoming disorganised. The sequence of interactions between system components during coordinated pressing and rest defence is illustrated in Figure 3.3.

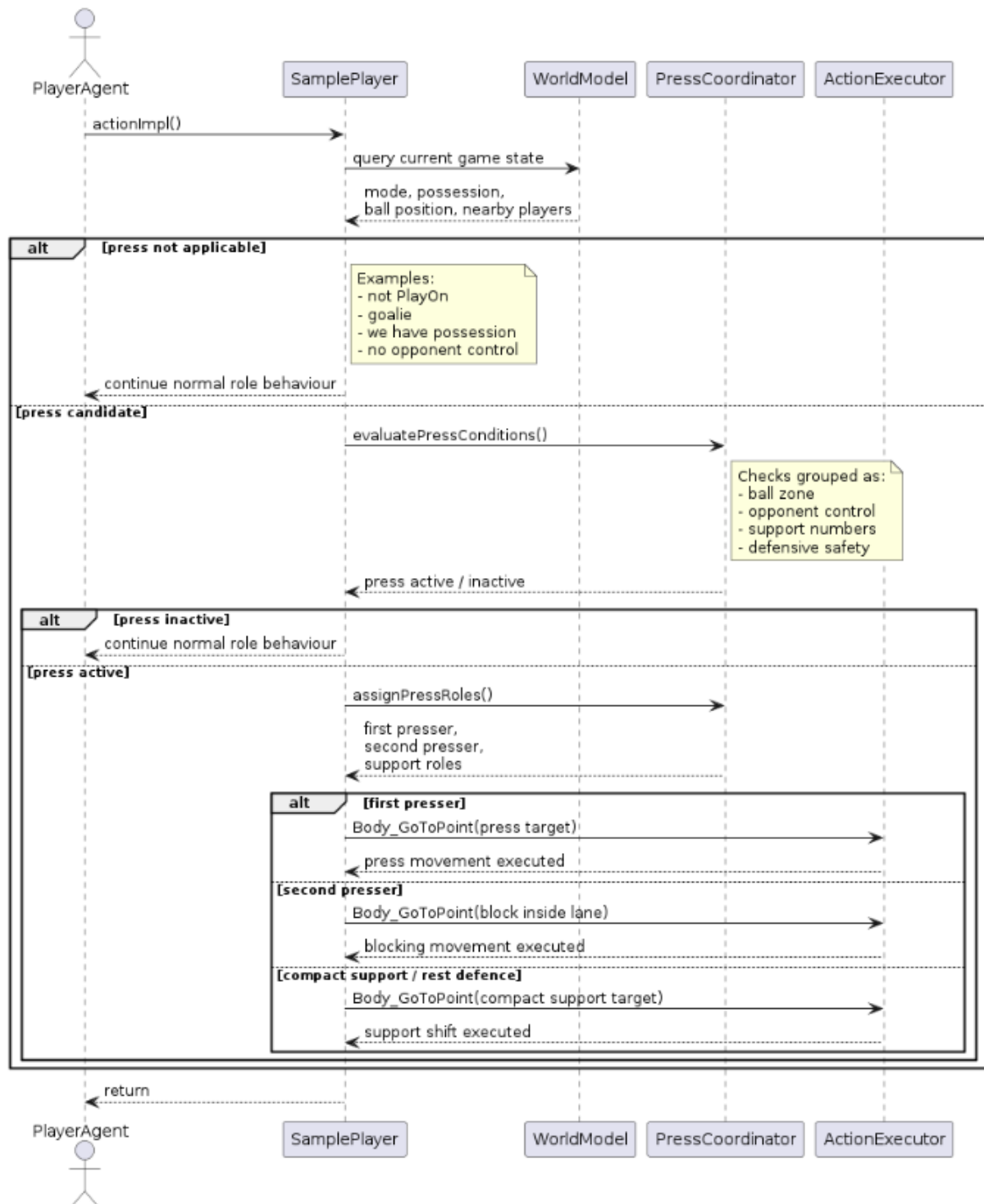


Figure 3.3: Sequence diagram illustrating coordinated pressing and rest defence activation across system components.

3.5 Goalkeeper Tactical Design (UC3, UC4)

The goalkeeper design introduces two behaviours, predictive positioning and support during buildup play.

Predictive positioning aims to reduce goals conceded by adjusting positioning based on ball position, opponent proximity and movement patterns, rather than solely reacting to the current ball state. The goalkeeper proactively moves to maximise the coverage of likely shooting angles.

Support behaviour allows the goalkeeper to leave his net during controlled phases of possession. This contributes to the buildup play, as it adds another passing option to help support ball retention. However, this behaviour has been constrained to ensure that goalkeeper responsibilities are not compromised.

Together these behaviours allow the goalkeeper to adapt dynamically between offensive support and defensive protection. The decision making process for goalkeeper behaviour selection is illustrated in Figure 3.4, while an example of goalkeeper support positioning during buildup play is shown in Figure 3.5.

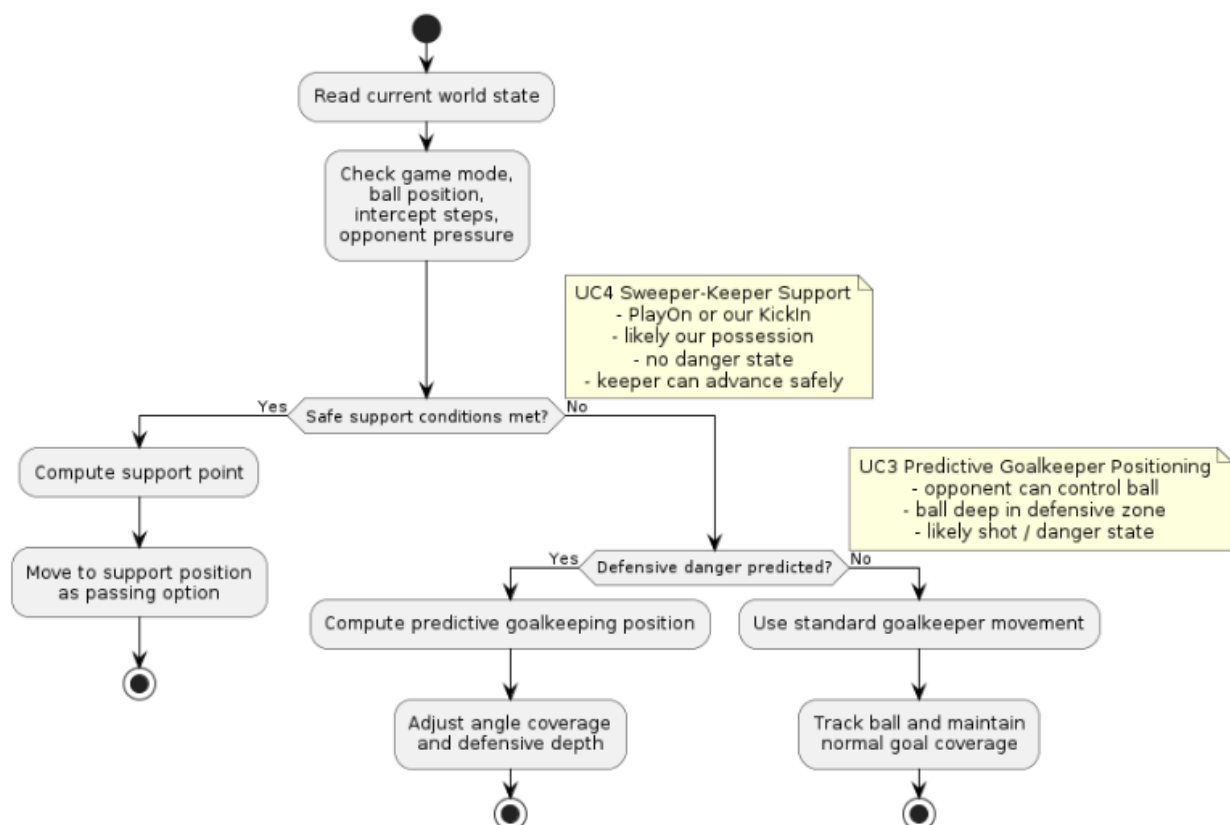


Figure 3.4: Goalkeeper decision making process for predictive positioning and support behaviour.

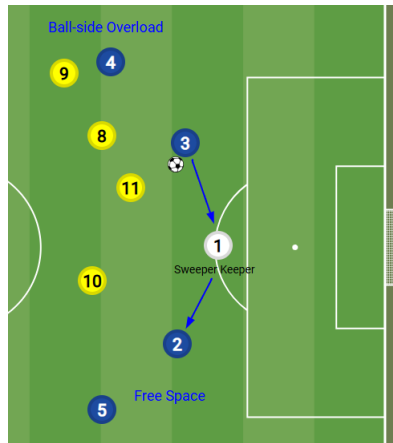


Figure 3.5: Example of goalkeeper support positioning during buildup play.

3.6 Adaptive CAM Behaviour Design (UC8)

The central attacking midfielder (CAM) is designed to be a context aware decision making agent that adapts behaviour based on the environmental conditions.

The CAM evaluates multiple different decisions, such as shooting, passing, dribbling and recycling possession. The move chosen to be executed is decided based on factors such as opponent pressure, space availability, the quality of receiving options and the current field position.

The CAM uses a rule based decision model which allows its actions to stay consistent with real football behaviours, this includes creating attacking opportunities and maintaining possession when under pressure. The decision making process of the CAM, including action selection under different contextual conditions, is illustrated in Figure 3.6.

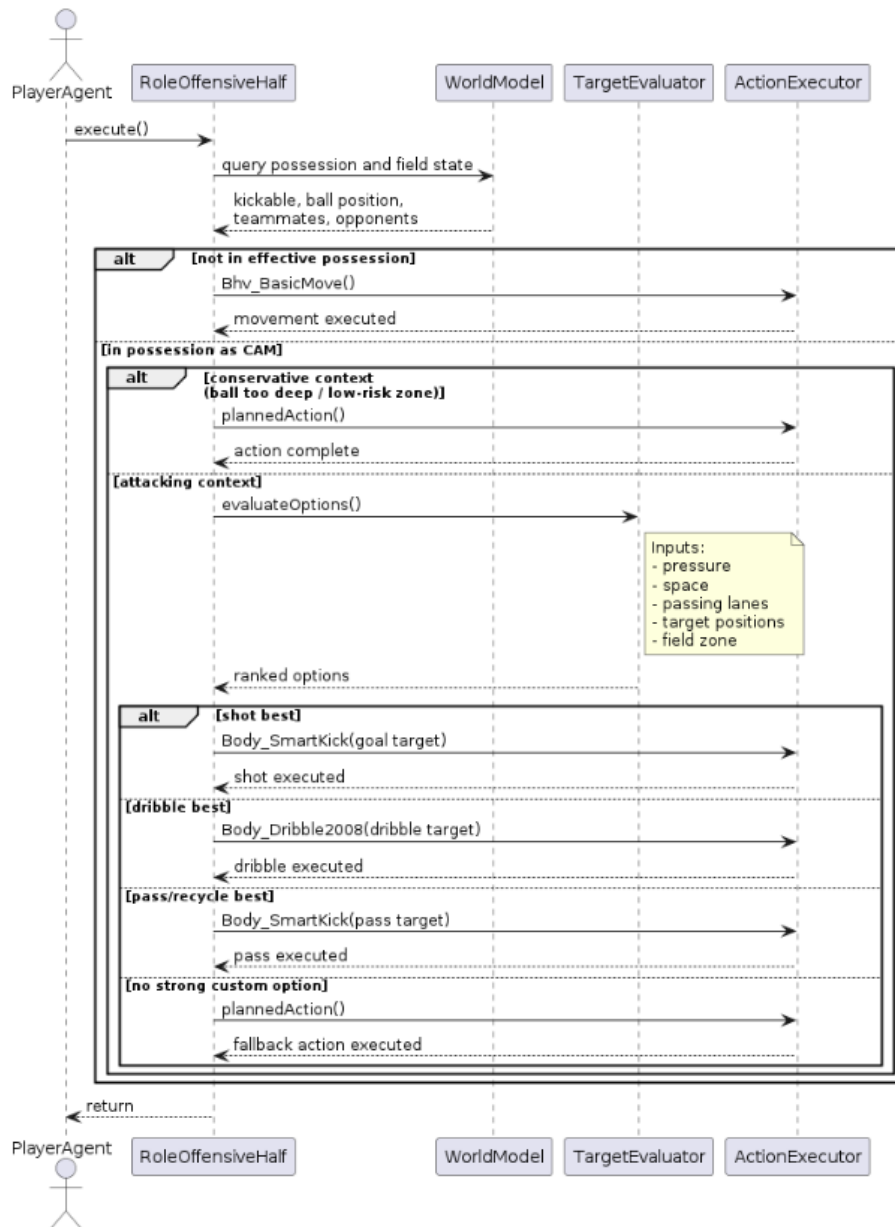


Figure 3.6: Sequence diagram illustrating adaptive CAM decision making across different game contexts and action selection pathways.

3.7 Wide Final Third and Cutback Design (UC9)

The wide attacking screen focuses on improving chance creation through structured behaviour in the final third. When the ball enters the final third in a wide area the system evaluates numerous potential passing options such as crossing, cut back, shooting or dribbling.

This is supported by coordinated movement from other players who will make runs to areas in alignment with where the wide attacker is planning to pass the ball to. This allows the wide attacker, when in a dangerous area to have meaningful passing options. The decision making process for wide final third actions are illustrated in Figure 3.7, while an example of coordinated attacking runs and cutback positioning is shown in Figure 3.8.

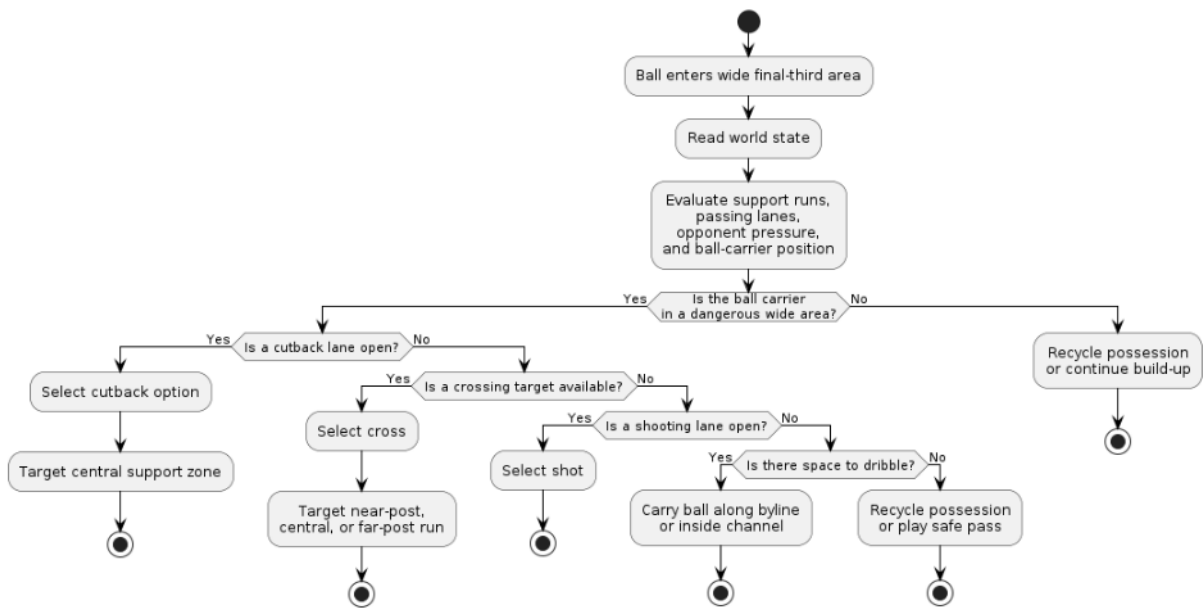


Figure 3.7: Decision making process for wide final third actions, including cutback selection, crossing, shooting, and ball recycling decisions.

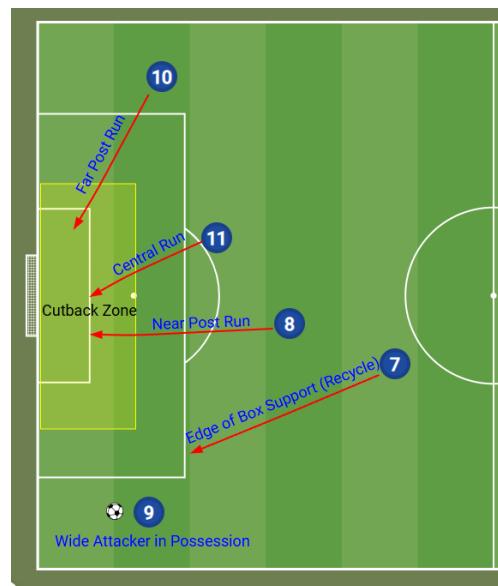


Figure 3.8: Example of coordinated attacking runs and cutback positioning showing central, near-post, and far-post movement patterns.

3.8 Set Piece Design (UC2)

Set pieces have been designed based on the current game state. The system has a range of set pieces including attacking and defensive free kicks, corners, kick-ins and kick-offs. For each set piece, different factors are assessed such as opponent positioning, the available passing options and the current ball position, applying the best balance between risk and possession control for the given scenario. This ensures that set pieces contribute positively to the overall performance rather than negatively impacting possession. The decision making process for set piece execution under different game conditions is illustrated in Figure 3.9.

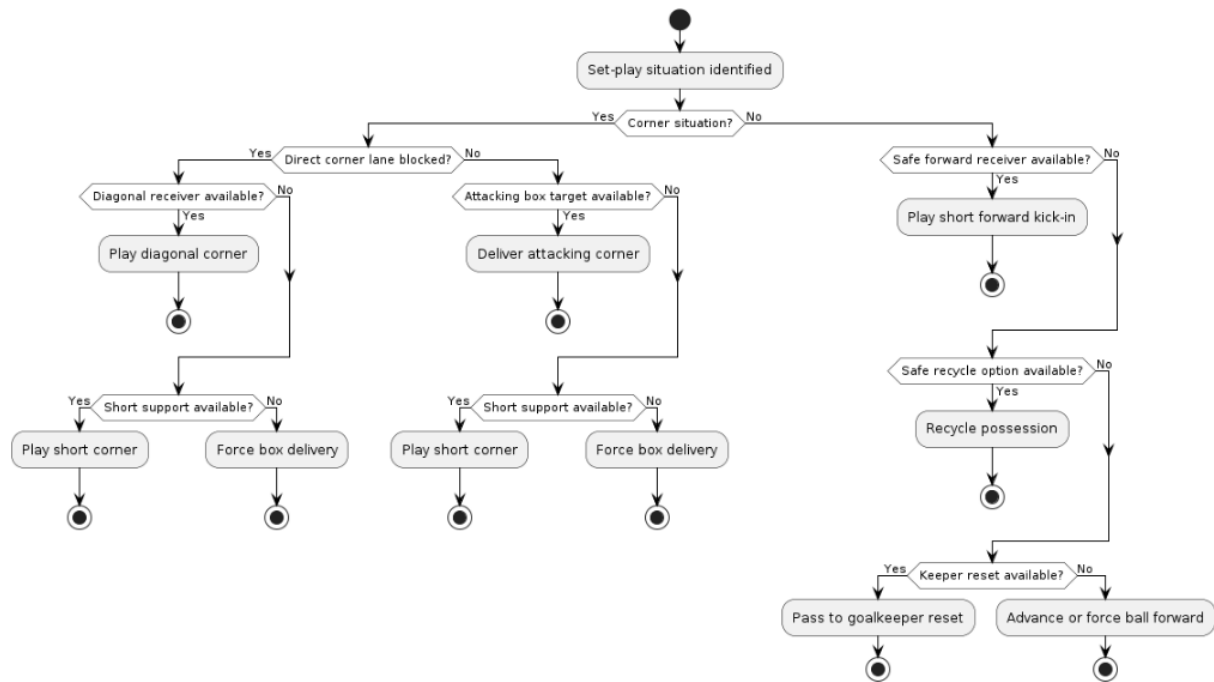


Figure 3.9: Decision making process for set piece execution, including corner, kick-in, and recycling options based on game context.

3.9 Match Analysis and Data Processing Design (UC10)

The system includes statistics and visualisations to aid the overall evaluation of the implemented tactics. The pipeline structure follows the order of data collection, parsing, analysing and finally, presenting outputs. The match data is extracted from the log files provided by the server and is transformed into graphs, heatmaps and figures. This allows for the analysis of key performance metrics such as possession, territorial control and individual agent performance. The match analysis and data processing pipeline is illustrated in Figure 3.10.

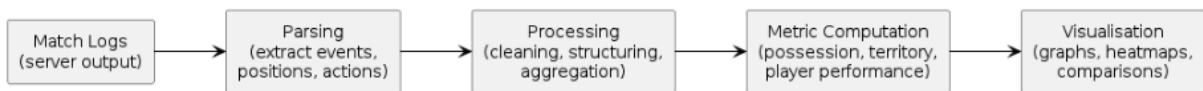


Figure 3.10: Match analysis and data processing pipeline from log extraction to visualisation of performance metrics.

3.10 Key Design Decisions, Trade-offs, and Limitations

Throughout the design process numerous factors were taken into consideration. Firstly, a rule based approach was preferred instead of machine learning, as this facilitated control of agent behaviours in certain situations, along with building in real football logic. Whilst this limits the systems adaptability, it allows for the tactics to be explicitly designed and analysed.

In place of a full redesign, a gradual extension of the existing system was chosen instead. This reduced the complexity whilst maintaining system stability, however this introduced limitations due to the reliance on the existing structures.

Another important decision was to allow players to adapt their positions instead of sticking to the starting formation due to the dynamic nature of football. Also, many behaviours are designed to

balance attacking intent with defensive stability, which represents a key trade off between managing risk and control while improving system performance. All agents in the RoboCup environment operate under uniform physical limitation including speed and stamina. This ensured the behaviours developed relied on tactical intelligence rather than physical advantages.

Finally, due to the RoboCup simulator producing primarily low-level data, the match statistics and visualisations have a slight degree of approximation, however it is still suitable for comparative analysis.

4.0 Implementation

This section describes the implementation of the system built on top of the HELIOS Base team. It outlines the technologies used and how the new tactical behaviours were implemented within the existing architecture.

4.1 Development Environment and Technologies

The core system was developed in C++. HELIOS Base was developed in this language, enabling direct integration with the existing architecture and allowing modification of player behaviours and tactical logic.

Development was carried out in a Linux based environment using Ubuntu via Windows Subsystem for Linux (WSL). Visual Studio Code was used as the environment for development allowing for editing, building and debugging of the system. The system was compiled and executed using Makefiles, consistent with the HELIOS build system.

The RoboCup simulation environment has multiple components, such as the simulation server and visualisation tools. The server manages the simulation including the agent's interactions and game management, whereas the monitor provides real time visualisation of the gameplay. These tools were used throughout the development of the project, to observe the changes in agent behaviour and analyse the impact of the tactics developed.

Python was also used in the project for creating a supporting analysis component which processes match data and produces visualisations. This included extracting key metrics such as goals scored and producing graphs to support the evaluation of the implemented tactics. Python was chosen due to its powerful libraries for data parsing, analysis and visualisation which makes it ideal for processing the RoboCup 2D logs and generating clear outputs. Version control was managed using GitLab, and the full implementation of the system is available in the project repository [1].

4.2 Integration with the HELIOS codebase

The most important implementation task was integrating new tactical logic into an already existing and complex code base. This required analysis of the system in order to identify the correct places to

implement the new behaviours within the HELIOS architecture, allowing new decision making logic to be evaluated against the original system.

The integration was achieved by adding logic to key decision points in the agents control process, such as movement, action selection and positioning whilst keeping the team stable.

4.3 Implementation of formation and positional structures

The formation was implemented using the HELIOS tool SoccerWindow2 alongside position layer updates in strategy.cpp. The intended 4-2-3-1 formation with an asymmetric double pivot was achieved by modifying each agent's role position, particularly player 7 where he was positioned ahead of player 6.

Positioning was also modified for wide attacking phases where offensive players dynamically adjust to improve support for the agent in possession. This includes far post runs, late central runs and recycle options. The implemented 4-2-3-1 formation and positional structure within the HELIOS environment is shown in Figure 4.1.

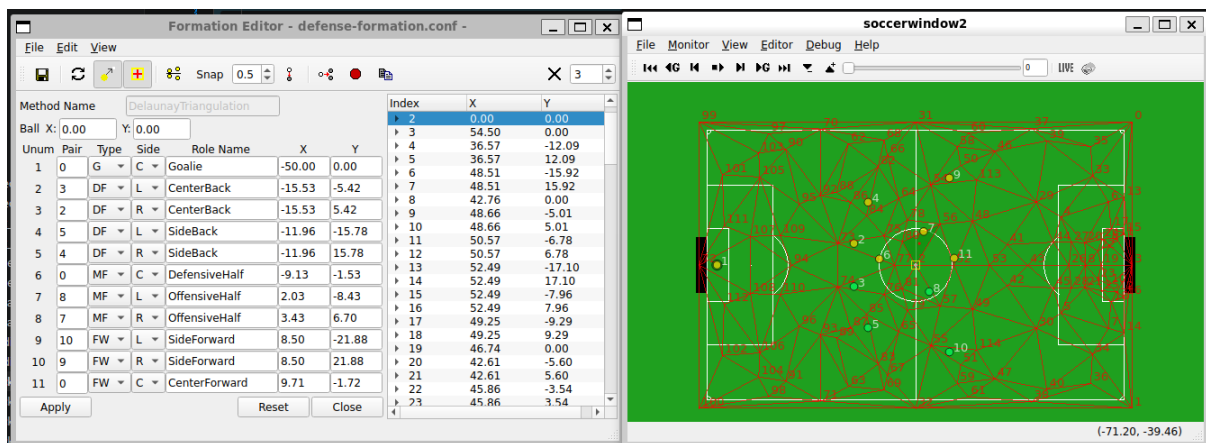


Figure 4.1: Ryan_FC formation configuration and SoccerWindow2 visualisation showing the implemented 4-2-3-1 structure with Delaunay triangulation.

4.4 Implementation of pressing and rest defence

Pressing behaviour was implemented across both bhv_basic_move.cpp and sample_player.cpp.

In bhv_basic_move.cpp, the conditions for attempting interceptions were modified to take into account roles and field position. Midfielders and wide defenders were allowed to attempt more aggressive interceptions in certain areas of the pitch.

A pressing system was implemented in sample_player.cpp through the doCoordinatedHighPress() function. This function decides when the players should press together based on conditions including ball location, support options, likelihood of opponent in possession. When the conditions are satisfied, the function assigns set pressing roles and coordinates their movement, such as striker shadow pressing, midfield screening and compact support behaviour.

Rest defence was implemented to maintain defensive stability when pressing. In `bhv_basic_move.cpp`, a blocking system predicts potential dribbling paths and assigns an appropriate player to cover them. In `sample_player.cpp`, additional players are positioned into a compact defensive structure based on available support and defensive line positioning. Together these ensure that defensive stability is balanced with aggressive pressing.

4.5 Implementation of goalkeeper behaviour

Goalkeeper behaviour was implemented in `bhv_goalie_basic_move.cpp` by adding two new behaviours.

The first implementation is sweeper keeper support. The function `shouldGoalieSupport()` evaluates whether the goalkeeper can safely advance as a passing option based on factors including threat level and current ball position. When activated, `getGoalieSupportPoint()` computes a support position behind the ball, and movement is executed using `Body_GoToPoint()`.

The second behaviour is predictive positioning. The function `shouldUsePredictiveGoalie()` identifies high risk scenarios, and `getPredictiveGoaliePos()` computes a position that maximises goal coverage based on likely shooting trajectories.

4.6 Implementation of adaptive CAM behaviour

The CAM behaviour was implemented in `role_offensive_half.cpp` by extending the `doKick()` function for player 8.

A set of helper functions was introduced to evaluate the current game context, including opponent pressure, passing options, and player positioning. These are used to generate and score multiple action choices, such as through passes, slip passes, wide distribution, recycling, and dribbling.

The action selection is implemented through a prioritised decision structure. Shooting actions in high value zones are evaluated first, followed by safe possession options under pressure, progressive passing options when beneficial, and dribbling when space is available.

4.7 Implementation of wide attacking and cutback behaviour

Wide attacking behaviour was implemented in `role_side_forward.cpp` within the winger's `doKick()` function. When the ball enters wide attacking zones, the agent uses a specialised decision making process.

The implementation identifies possible passing options in key areas such as the near post, far post, penalty spot and the edge of the box. Helper functions evaluate each function based on factors such as opponent proximity. A list of possible actions have been implemented which includes cutbacks, shots, inside passes, recycling and dribbling. These actions are then given a score and the highest scored options are then prioritised.

Supporting player positions were implemented in the strategy layer ensuring team wide levels of coordination in attacks.

4.8 Implementation of set piece routines

Set piece behaviour was implemented across `bhv_set_play_free_kick.cpp`, `bhv_set_play_kick_in.cpp` and `bhv_set_play_kick_off.cpp`.

Free kicks are classified into defensive, middle third, and attacking scenarios. Receiver selection is implemented using scoring functions that evaluate spacing, forward progression, and pass lane safety. The system evaluates multiple options for kick-ins and corners, such as attacking deliveries, short passes and recycling, selecting the best available option that balances offensive progression with possession control. Finally for kick-offs they execute a controlled back pass to a centre-back.

4.9 Implementation of match analysis and visualisation

Analysis metrics and visualisations were implemented using Bash and Python. Bash scripts automate batch match execution, including server initialisation, team launch, process management, and log collection.

The first Python script processes `.rcg` files by reconstructing match state from simulation data, including ball position, player state, and play modes. From this, key metrics such as possession, passes, goals, assists, and player movement are calculated and stored in a structured JSON format. The second python script collates data together across multiple matches and computes overall statistics and generates outputs such as heat maps, formation summaries and match reports.

4.10 Implementation challenges

Several implementation challenges were encountered. Integration with an established code base required analysis of the system, to select appropriate modification points to ensure compatibility with the original baseline team. Implementing the modifications was challenging, due to dynamic agent behaviour. They do not stay perfectly organised and many of the implemented tactics had threshold based parameters which required iterative tuning. The final implementation challenge was collating reliable statistics and outputs due to the RCG files only containing raw data.

5.0 Testing

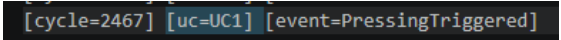
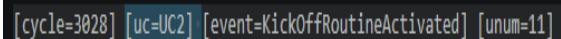
5.1 Test Plan

The system was tested by running matches against the HELIOS Base team in the RoboCup 2D Soccer Simulator environment. Since the simulation is dynamic, it wasn't possible to create controlled and repeatable tests. Instead, testing was carried out by observing full matches under different conditions, generated by the server and its stochastic nature to see whether the team and player behaviours

matched the expected outcomes. To further confirm, outputs were printed when the system executed the new implemented behaviours.

165 matches were used for testing. This included 150 games which were ran at 3 games a time at maximum speed as well as a further 15 games which were ran with the monitor enabled. This ensured there were both large scale analysis and also detailed agent analysis. These simulated matches also allowed for the non-functional requirements to be tested including performance, stability and consistency.

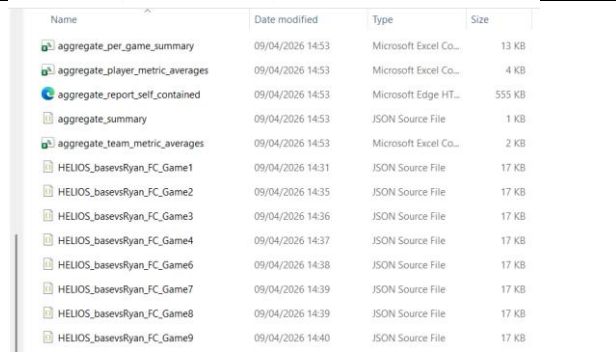
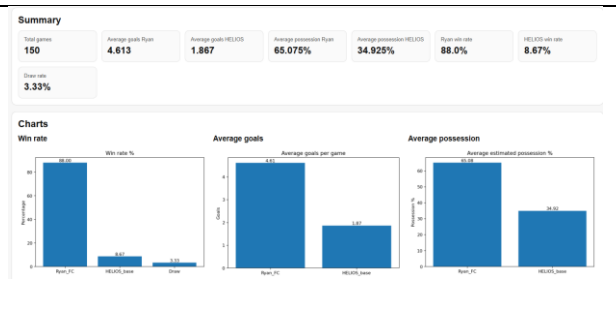
5.2 Test Results

Test ID	Feature	Preconditions	Expected Outcome	Result	Evidence
1.0	UC 1 Pressing: Press Trigger Activation	Press conditions met	The closest player to the ball initiates pressure by moving towards the ball carrier	Pass	RyanFC v Helios Game 15 - YouTube , (0:22) 
1.1	UC1 Pressing: Press coordination	Press is initiated	Multiple nearby players support the press by closing space and maintaining a compact structure	Pass	
1.2	UC 1 Pressing: Press Broken	Opposition escapes the press	The team reorganises into its defensive shape with players returning to position	Pass	RyanFC v Helios Game 13 , (5:21)
2.0	UC 2 Set Pieces: Kick-off routine	Team kick-off	The ball is played to a centre back to allow the team to establish shape	Pass	RyanFC v Helios Game 8 , (0:21) 

2.1	UC 2 Set Pieces: Free kick routine	Attacking free kick	Players position themselves in a structured formation and execute a coordinated action	Pass	RyanFC v Helios Game 12 , (3:26)
2.2	UC 2 Set Pieces: Kick- in routine	Team kick-in	The ball is played to a suitable nearby option while maintaining possession	Pass	RyanFC v Helios Game 1 - YouTube , 2:47 [cycle=1739] [uc=UC2] [event=ThrowInRoutineActivated] [unum=9]
2.3	UC 2 Set Pieces: Corner kick routine	Team corner kick	Players move into predefined attacking areas and a coordinated delivery is executed	Pass	RyanFC v Helios Game 7 , (4:30)
3.0	UC 3: Predictive shot stopper	Opposition attacking	The goalkeeper adjusts positioning based on ball location and attacking threat to improve shot coverage	Pass	Goalkeeper Movement [cycle=5045] [uc=UC3] [event=PredictiveGoalieActivated] [unum=1]

4.0	UC 4: Sweeper Keeper (Support Phase)	Team in controlled possession	The goalkeeper advances to provide an additional passing option	Pass	
4.1	UC 4: Sweeper Keeper (Defensive Phase)	Opposition attacking	The goalkeeper remains positioned near the goal and does not advance unnecessarily	Pass	
4.2	UC 4: Sweeper Keeper (Loose Ball)	Loose ball in defensive area	The goalkeeper moves to collect or clear the ball when appropriate	Pass	RyanFC v Helios Game 14 (2:15) <code>[cycle=515] [uc=UC4] [event=SweeperKeeperActivated] [unum=1]</code>
5.0	UC 5: Formation Structure (4- 2-3-1)	Game is in PlayOn mode	The team maintains a 4-2- 3-1 shape with appropriate positional structure	Pass	<code>[cycle=25] [uc=UC5] [event=Formation4231DefensiveUsed]</code> <code>[cycle=28] [uc=UC5] [event=Formation4231NormalUsed]</code> <code>[cycle=30] [uc=UC5] [event=Formation4231AttackingUsed]</code> 
6.0	UC 6: Double Pivot Behaviour	Game in play	One midfielder maintains a deeper position while the other advances to	Pass	<code>[cycle=30] [uc=UC6] [event=DoublePivotUsed]</code>

			support attacking play		
7.0	UC 7: Rest Defence	Team in attacking phase	Defensive players remain positioned to protect against counter attacks and maintain structure	Pass	 <pre>[cycle=75] [uc=UC7] [event=RestDefenceBlock] [unum=2]</pre>
8.0	UC 8: Adaptive CAM behaviour	CAM in possession	The player selects an appropriate action based on context rather than default behaviour	Pass	RyanFC v Helios Game 10: Slip pass (1:25), Recycle pass (2:59), Shot (3:16), Carry (6:17), Wide pass (10:24) <pre>[cycle=2084] [uc=UC8] [event=Carry] [unum=8] target=(4.19084,14) [cycle=2091] [uc=UC8] [event=WidePass] [unum=8] target=(10.3633,16.122) speed=1.58795 [cycle=3161] [uc=UC8] [event=SlipPass] [unum=8] target=(44.3622,-3.9636) speed=1.64136</pre>
9.0	UC 9: Wide Cutback Behaviour	Wide player in final third with central option available	A cutback pass is played into a central attacking area	Pass	RyanFC v Helios Game 3, (2:24) <pre>[cycle=1680] [uc=UC9] [event=EnteredWideFinalThird] [unum=9] [cycle=1680] [uc=UC9] [event=CutbackTargetFound] [unum=9] [cycle=1680] [uc=UC9] [event=Cutback] [unum=9]</pre>
9.1	UC 9: Wide Decision Adaptation	Wide player in final third with no viable cutback option	The player selects an alternative action such as passing, dribbling, or recycling possession	Pass	RyanFC v Helios Game 14 - YouTube Attack more space (3:56) RyanFC v Helios Game 10 Inside Pass (5:48), Shot (9:10) <pre>[cycle=1525] [uc=UC9] [event=AttackMoreSpace] [unum=10] self=(36.0520,13.8407) ball=(36.5874,13.3761) [cycle=1525] [uc=UC9] [event=AttackMoreSpace] [unum=10] target=(40.7874,10.3761) [cycle=3532] [uc=UC9] [event=EnteredWideFinalThird] [unum=9] [cycle=3532] [uc=UC9] [event=NoCutbackTarget] [unum=9] [cycle=3532] [uc=UC9] [event=BestAction_InsidePass] [unum=9] [cycle=5234] [uc=UC9] [event=BestAction_Recycle] [unum=9] [cycle=4143] [uc=UC9] [event=Shot] [unum=9] target=(52.5,7.5) speed=2.95</pre>

10.0	UC 10: Data Processing	Match log files available	The system processes match data without errors and extracts relevant performance information	Pass	
10.1	UC 10: Visualisation Output	Processed match data available	The system generates visual outputs representing key performance metrics	Pass	

5.3 Testing Summary

During the development phase, several issues were encountered which caused some of the use cases to not meet the expected outcomes. For example, UC4 sweeper keeper early implementations caused the goalkeeper to be too aggressive and not recover in time, leading to goals being conceded. Each issue was identified and resolved through testing and refinements, which has led to the final system passing all functional tests, with the correct behaviours being displayed. The 165 matches showed that the system was able to meet all the non-functional requirements, as it operated without any run time errors or unexpected behaviour. Overall, these tests give confidence that the system works as expected and meets the requirements.

6.0 System Evaluation and Experimental Results

6.1 Evaluation Methodology

The evaluation of this project was to determine whether the newly developed RoboCup Team (Ryan_FC), produced measurable improvements over the original HELIOS Base team in which it was expanded from. The improvements are based on both match performance, as well as the successful implementation of a new playing identity based on the tactical behaviours developed. The evaluation was designed to assess if the modified team produced better match outcomes, but also whether the tactics added to the team were visible and consistent.

To support the evaluation, a baseline comparison was created where HELIOS Base played simulated matches against itself. This was then compared against simulated matches between Ryan_FC and

HELIOS Base. The baseline experiment was used to analyse the behaviour and performance of the unmodified team within the simulation environment. The experimental setup was used to evaluate the impact of the tactical and behavioural changes to the team.

The evaluation was conducted across two levels. This consisted of both quantitative and qualitative analysis.

Quantitative Evaluation

For the quantitative evaluation, 150 matches were run for each experiment without the rcssmonitor. This resulted in:

- 150 HELIOS Base vs HELIOS Base matches
- 150 Ryan_FC vs HELIOS Base matches

These matches were used to collect data, combine results and produce meaningful information. The primary metrics collected included:

- Win rate
- Draw rate
- Lose rate
- Goals scored
- Goals conceded
- Goal difference
- Estimated possession

150 games were chosen for each experiment for two reasons. Firstly, the RoboCup 2D simulation environment naturally produces random elements and secondly, the agents interact with each other in complex ways. This means that even slight changes can lead to very different outcomes, hence evaluating performance over a limited number of games could produce a misleading conclusion caused by outlier results. Running 150 games reduced the impact of random variation and this resulted in a more reliable set of results for comparison.

Qualitative Evaluation

A further 15 matches were executed for each condition, this time the rcssmonitor was enabled and the games were recorded for detailed analysis. This resulted in:

- 15 monitored HELIOS Base vs HELIOS Base matches
- 15 monitored Ryan_FC vs HELIOS Base matches

The recorded matches were used to evaluate whether the tactical behaviours stated in the use cases were visibly apparent in the games, alongside determining each behaviour's individual effectiveness within the simulation. Since the 150 game batch provided statistical reliability, the monitored matches

allowed for a different type of evaluation, where the tactics developed could be inspected separately and visually, which the quantitative evaluation was unable to do.

These matches were analysed both visually and through a custom analysis script, which generated additional metrics and visualisations, including:

- Team heatmaps
- Player heatmaps
- Possession distribution
- Ball cycles in own half, opposition half, final third and penalty area
- Averaged visual summaries
- Player level attacking and passing outputs

The outputs were developed as support evaluation for the specific implemented tactics such as pressing, rest defence, adaptive CAM behaviour and structural differences between Ryan_FC and HELIOS Base.

Definition of Improved Criteria

Improvement has been evaluated using both quantitative and qualitative criteria.

Quantitative criteria:

- Higher win rate compared to the baseline experiment
- Higher average goals scored
- Lower average goals conceded
- Greater average goal difference
- Increased overall possession
- Increased territorial control

Qualitative criteria:

- Consistent execution of the developed tactics
- Structural patterns aligned with the designed system
- Clear behavioural differences between Ryan_FC and HELIOS Base

Using both quantitative and qualitative improvement criteria ensures that the evaluation didn't focus solely on match outcomes, but also the overall success of the development compared to the design objectives.

Evaluation Rationale

The evaluation approach integrated both statistical analysis and detailed behavioural analysis. The 150 game experiment provided statistically meaningful evidence of Ryan_FC's overall performance,

whereas the 15 recorded games provided tactical and behavioural evidence both statistically and visibly, to show how the results achieved actually happened. This approach ensured that the evaluation was fair, objective and comprehensive. 330 matches were run which provided a strong basis for assessing the effectiveness and consistency of the tactics developed for Ryan_FC.

6.2 Quantitative Results

The quantitative evaluation compared results of Ryan_FC against the original HELIOS Base team across 150 matches. These results were evaluated against 150 matches of HELIOS Base vs Helios Base which acted as a baseline. The baseline condition acted as a benchmark for performance of the unmodified team under identical simulation conditions.

Baseline Results: HELIOS Base vs Helios Base

The baseline experiment produced balanced results across the 150 simulated games. HELIOS2 achieved a win rate of 43.33%, while HELIOS1 achieved a win rate of 50.67% and overall, there was a 6.0% draw rate. The number of goals scored was almost equal as HELIOS2 scored 413 goals and HELIOS1 scored 428 goals. Possession was also balanced, 50.459% for HELIOS2 and 49.541% for HELIOS1, as shown in Figure 6.1. Spatial control was also balanced between both teams, with no consistent dominance observed in advanced areas of the pitch. The average goal difference across the 150 matches was approximately 0.1. These statistics indicate that there was no significant advantage for either team, this confirms that the HELIOS Base system is internally balanced and can be used as a reference point when judging improvements.

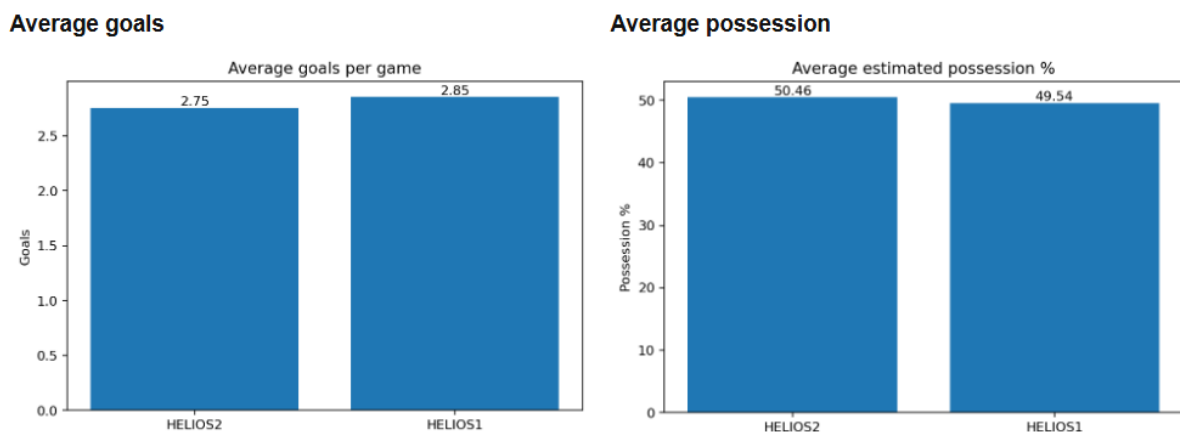


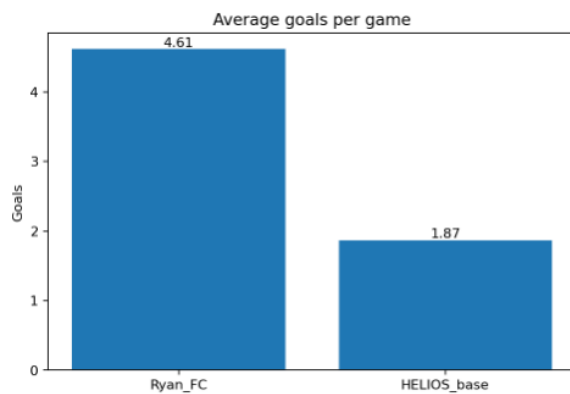
Figure 6.1: Baseline performance comparison between HELIOS2 and HELIOS1: (a) average goals per game; (b) average possession (%).

Experiment Results: Ryan_FC vs HELIOS Base

The 150 games of Ryan_FC vs HELIOS Base shows a substantial improvement in performance, Ryan_FC achieved a win rate of 88%, whereas HELIOS only achieved a 8.67% win rate and a 3.3% draw rate.

Ryan_FC scored an average of 4.61 goals per game, whilst HELIOS average goals per game was 1.87. Ryan_FC also heavily dominated possession at 65.08% compared to 34.92% for HELIOS, as shown in Figure 6.2. Ryan_FC demonstrated clear territorial dominance, with the ball consistently being retained in more advanced areas such as the opponent half, final third, and penalty area. The average goal difference was 2.747. These stats indicate a clear and consistent advantage in performance for Ryan_FC and represents a significant improvement over the baseline results (Figure 6.1), where both teams demonstrated near identical performance across all metrics.

Average goals



Average possession

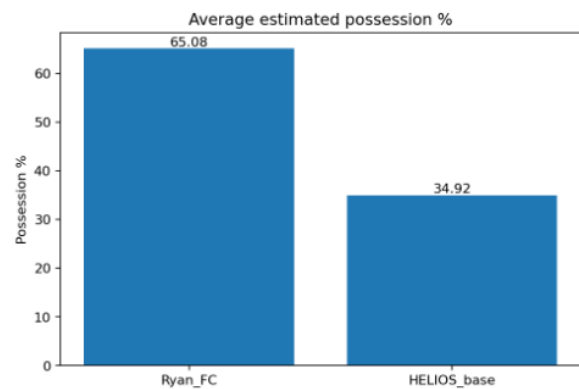


Figure 6.2: Performance comparison between Ryan_FC and HELIOS Base: (a) average goals per game; (b) average possession (%).

Comparative Summary

Table 6.1: Comparative summary of baseline and experimental results across key performance metrics.

Matches	Team	Games	Win Rate	Draw Rate	Total Goals	Avg Goals Scored	Avg Goals Conceded	Avg Possession	Avg Goal Difference
HELIOS vs HELIOS	HELIOS1	150	43.33%	6.00%	428	2.753	2.853	50.46%	-0.1
HELIOS vs HELIOS	HELIOS2	150	50.67%	6.00%	413	2.853	2.753	49.54%	0.1
Ryan_FC vs HELIOS	Ryan_FC	150	88.00%	3.33%	692	4.613	1.867	65.08%	2.747
Ryan_FC vs HELIOS	HELIOS	150	8.67%	3.33%	280	1.867	4.613	34.93%	-2.747

The contrast between the two experiments is significant, as shown in Table 6.1. In the baseline, both teams remained close across all the stats, which suggests that unmodified HELIOS teams are evenly matched. However, Ryan_FC displayed a substantial performance increase, as the team achieved a higher win rate, better attacking output and greater possession dominance. Ryan_FC scored approximately 1.76 more goals per game compared to both the HELIOS teams in the baseline experiment, whilst also conceding less goals than the baseline average.

Quantitative Interpretation

The quantitative results strongly suggest that the tactical behaviours implemented improved overall team performance. Ryan_FC outperformed HELIOS across all major metrics, as the increase in win rate from a balanced baseline (~50%) to 88% demonstrates a clear competitive advantage. This is supported by the average number of goals scored, as it increased from approximately 2.8 in the baseline to 4.6 for Ryan_FC. In addition, goals conceded was reduced, suggesting improved defensive stability.

The increase in possession from approximately 50% to 65% and the increase in spatial control, further implies that Ryan_FC has a greater control over matches and consistently maintains the ball in advanced areas of the pitch.

To conclude, these results indicate that the implemented tactical and behavioural changes do not only improve match outcomes, but also enhance control of possession, space, and attacking opportunities within the simulation.

6.3 Qualitative Evaluation

The 15 game recorded batch was used to evaluate whether the implemented tactics were visibly present during a match and whether they appeared to contribute to the improved performance, as seen in section 6.2. Since the 150 games gave reliable statistical performance, the recorded games instead allow for closer inspection of player roles, decision making and team structure through the recorded gameplay, heatmaps, player level outputs and formation visuals.

Similar results were achieved in this 30 match evaluation, with the baseline 15 games remaining broadly balanced. Possession was split at 48.72% and 51.28%. Territorial control was also balanced with the ball spending approximately an even amount of time in each team's half (2274.8 and 2413.8 cycles) and in each team's penalty box (272.0 and 282.87 cycles). The overall goal difference was also minimal, with only a 4 goal difference after the 15 games. In contrast, Ryan_FC showed clear dominance in the games again, as they had an average of 65.57% possession. The ball spent considerably more time in the HELIOS Base side of the pitch, with the ball having a total time in their half of 2678.67 cycles. The ball spent 1457.53 cycles in their final third and 259.33 cycles the ball was in their penalty box for. The ball spent marginally less time in Ryan_FCs half of the pitch. Only 1618.27 cycles the ball was in Ryan_FCs half, 715.8 cycles in their final third, and 124.87 in their penalty area. Total goal difference was massively increased as well with Ryan_FC scoring 78 goals compared to HELIOS Base's 24 goals. This indicates that the implemented tactics were not only visible but contributed to Ryan_FC having stronger spatial control, more sustained attacking pressure and better defending structure.

Team Control and Defensive Structure

The most immediate difference between the baseline and Ryan_FC's 15 game experiment was the overall control Ryan_FC has in the matches. The improvement appears as a result of several behaviours collaboratively, such as the pressing behaviour (UC1), the 4-2-3-1 formation (UC5), the double pivot (UC6) and rest defence (UC7).

In the baseline HELIOS vs HELIOS matches, the teams remained evenly balanced in possession, territory and scoreline, which was expected. However, in the Ryan_FC vs HELIOS games, the territorial differences were massive. Ryan_FC averaged 2678.67 ball cycles in the opponent half compared to 1618.27 for HELIOS Base, whilst more than doubling how long the ball spent in the HELIOS Base's final third, compared to how long it spent in their own final third (1457.53 to 715.8 cycles). This suggests that Ryan_FC kept the ball in higher areas of the pitch and were able to attack for longer periods, which in turn did not allow HELIOS to control where the game was being played in terms of field position. This shift in territorial control is illustrated in Figure 6.3, where Ryan_FC shows a significantly higher proportion of ball cycles in the opponent half and final third, while HELIOS_base retains more possession within its own half.

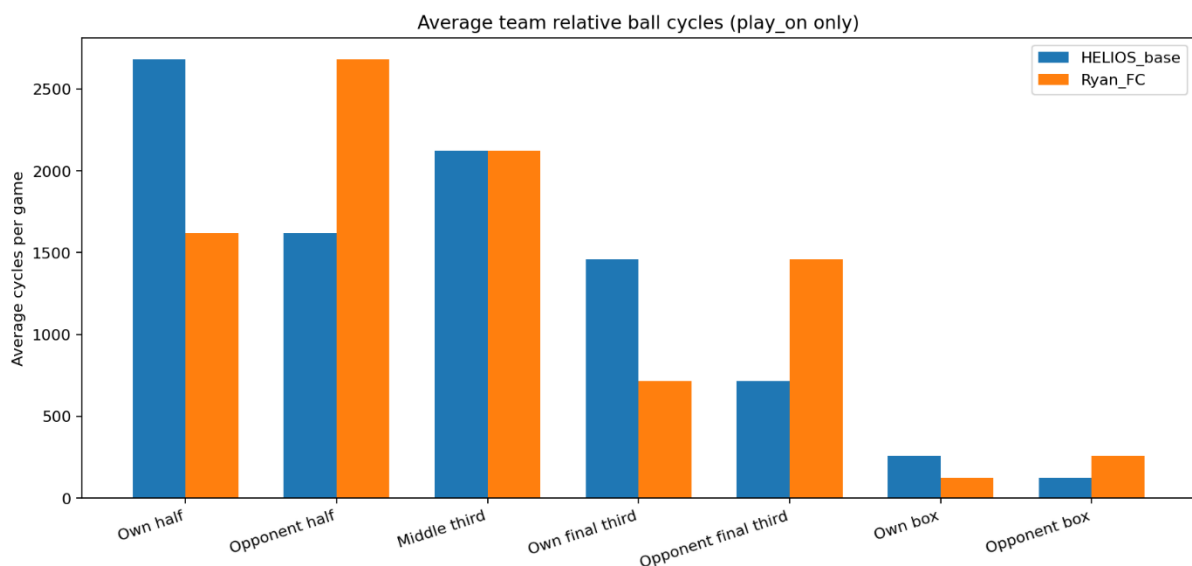


Figure 6.3: Average team ball cycles across pitch regions for Ryan_FC and HELIOS Base.

The improved control is consistent with the intended effect of the pressing and rest defence behaviour (UCs 1 and 7 respectively). Pressing helped Ryan_FC to regain and maintain the control of the game, whereas the rest defence reduced the risk of losing control immediately by committing too many players forward. These tactics allowed Ryan_FC to be aggressive in winning the ball but also protected the chances of counter attacks, this is backed by the match outcomes where Ryan_FC averaged 65.57% of the possession, scored an average of 5.1 goals and conceded an average of only 1.6, whilst HELIOS Base produce the reverse stats (34.43% of possession, 1.6 goals per game average and 5.1 goals

conceded per game). These two tactics show that Ryan_FC were able to control the game and create chances while remaining defensively stable.

The average formation outputs show that both teams kept their shape properly throughout the game. However, Ryan_FC operated higher up the pitch compared to HELIOS Base, which reinforces that the improvement in possession and territorial control is due to behavioural differences, such as pressing and rest defence.

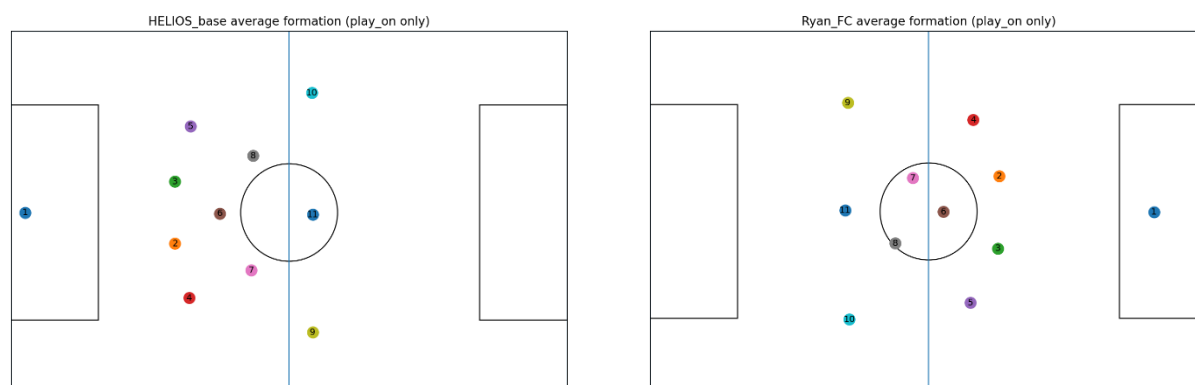


Figure 6.4: Average formation positions (play-on only) for HELIOS Base (left) and Ryan_FC (right).

The asymmetric double pivot (UC6) also appears to have contributed to the overall control of the games. As seen in the average formation diagram (Figure 6.4), player 7 does occupy a position between the deeper defensive midfielder (6) and the more offensive attacking midfielder (8), indicating the presence of the asymmetrical pivot structure. Upon analysis of player level data, player 7 averaged 10.93 passes a game with an 85.98% pass completion rate. Player 7 also has an estimated average of 0.6 goals assists a game, with a total of 9 assists in the 15 games. Player 7 registered 2 goals throughout the 15 matches and covered an average distance of 1883.59 simulation units per match. These stats and average formation positioning suggested that player 7 inherited a transitional linking role, where they were involved in both attacking and defending, contributing to ball progression and chance creation while maintaining defensive balance, leading to an increased performance by Ryan_FC.

Overall, the recorded matches strongly support that the implemented structural and defensive use cases did work together to make the team more dominant and defensively solid.

Goalkeeper behaviour

The monitored matches were used to assess the predictive goalkeeper (UC3) and sweeper keeper behaviours (UC4). These use cases are less directly measurable through overall team statistics, so the evaluation relied more on visual observation.

The predictive goalkeeper behaviour (UC3) aimed to improve save percentage by predictively moving based on the danger, rather than reacting after the shot. The monitored recordings of matches allowed for this behaviour to be reviewed manually, and observations were made based on how Ryan_FC's

keeper moved when the ball was advancing near the goal, in comparison to how HELIOS Base goalkeeper in the same scenarios.

The sweeper keeper behaviour (UC4) was analysed in the same way, by manually reviewing the recorded games to identify if RyanFC's keeper became an extra passing option during build up play and to review if the goalkeeper collected loose balls when safe. As well as the recordings, the script provided useful visualisations of the goalkeeper's movement. The goalkeeper heatmaps below shows that Ryan_FC's keeper throughout the games occupied a wider and more advanced area of activity, whereas the HELIOS Base keeper stayed relatively close to the net. The average formation output also displayed that Ryan_FC's keeper has a higher average position compared to HELIOS's. These outputs show characteristics of a sweeper keeper in terms of acting as a support option and increasing defensive coverage.



Figure 6.5: Goalkeeper positional heatmaps for HELIOS Base (left) and Ryan_FC (right), illustrating the increased advanced positioning of the Ryan_FC goalkeeper during match play.

The sweeper keeper behaviour was added for two reasons. It was added to improve defensive coverage, as well as to provide RyanFC with an extra passing option during build up play. The positional outputs suggest that this was successfully implemented, as the goalkeeper consistently occupied more advanced positions compared to HELIOS Bases. However, when visually analysing the game recordings, RyanFC never passed the ball back to the goalkeeper even when in an advanced position. Whilst the sweeper keeper logic was there, this suggests that the outfield agents didn't use it due to their decision making logic in terms of risk-based considerations.

Ryan_FC conceded 1.6 goals per recorded game and Ryan_FC's keeper recorded a better save percentage over the 15 games (47% compared to HELIOS 43%). This suggests that the updated goalkeeper logic didn't negatively impact the defensive performance of the team. This is important, considering an aggressive sweeper keeper can become a liability if not implemented carefully.

Overall, the outputs and recorded matches suggest that the predictive keeper (UC3) and sweeper keeper behaviours (UC4) were added successfully without negatively impacting overall team performance.

Attacking Patterns

The attacking behaviour of Ryan_FC reflects the implementation of attacking use cases, particularly adaptive CAM behaviour (UC8) and cutback based attacking patterns (UC9). The 15 game analysis showed clear structural differences compared to the baseline HELIOS system, especially in how the ball was progressed into advanced areas and converted into goal scoring opportunities

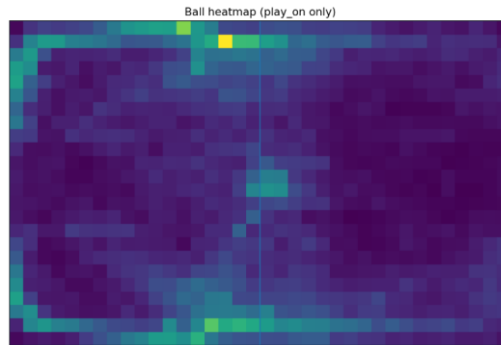


Figure 6.6: Average ball position heatmap (play-on only) for Ryan_FC, showing concentration of activity in wide attacking areas and central progression zones.

The average ball heatmap for Ryan_FC vs HELIOS (Figure 6.6) highlights a clear concentration of activity in the wide attacking channels, followed by movement into central zones. This pattern is consistent with the intended cutback behaviour (UC9), where the ball is progressed down the wings and then delivered into a central, high value scoring area. This is reinforced by the heat maps for the two Ryan_FC wingers (Figure 6.7), which illustrates consistent wide positioning followed by diagonal inward movement.

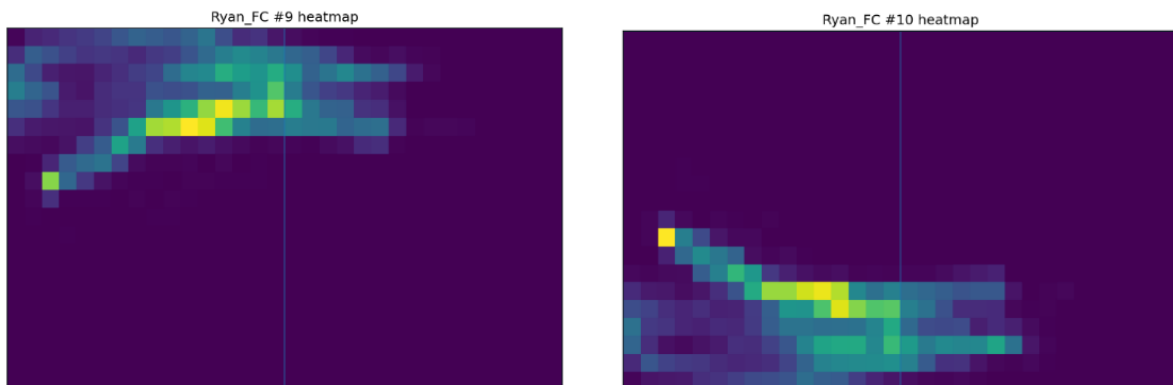


Figure 6.7: Positional heatmaps for Ryan_FC wingers (players 9 and 10), illustrating wide positioning and diagonal inward movement during attacking phases.

The role of the central attacking midfielder (player 8) is also important in linking these attacking phases. The heatmap for player 8 (Figure 6.8) shows a high concentration of activity in advanced central areas, particularly in the HELIOS's half and final third of the pitch. This positioning supports the behaviour expected by use case 8, where the player supports attacking phases dynamically and occupies space that allow them to receive inward passes from wide areas. Player 8 acts as central connector between wide build up play and final execution.

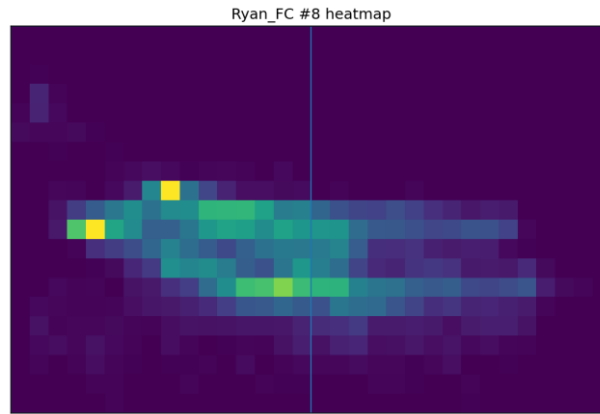


Figure 6.8: Positional heatmap for the central attacking midfielder (player 8), showing high activity in advanced central areas of the pitch.

The effectiveness of these attacking behaviours can be seen through the output metrics. Players 9 and 10 (wingers) averaged 1.87 and 1.33 estimated assists a game respectively. This suggests that wide attacking play frequently ended with a goal scoring opportunity. Player 8 also contributed to attacking output, averaging 0.4 assists per game and scored 4 goals in the 15 games. At a team level, the average number of cycles the ball was in the HELIOS's penalty area was 259.33, compared to only 124.87 cycles in Ryan_FC's penalty area. This means Ryan_FC was in an extremely dangerous attacking area for double the amount of time as HELIOS. Manual inspection of the 15 recorded matches identified that approximately 41 of Ryan_FC's goals involved wide progression by winger, followed by a pass into the centre of the box. Some of these actions may have been recycle or slip passes as their spatial characteristics closely resemble cut back behaviour. As a result, these should be interpreted as cutback consistent rather than definitively classified cutbacks. This further supports the implementation of UC9. Overall, the combination of positional outputs, statistics and observed goal patterns provide strong evidence that both the attacking use cases (UC8 adaptive CAM behaviour and UC9 cutback based attacking patterns) were successfully added and played a key role in improving RyanFC's attacking performance.

Set Piece Routines

Set piece behaviours (UC2) were evaluated primarily through monitored recordings rather than statistical analysis. The recorded matches allowed for inspection of the structured behaviour during corners, freekicks and kick-ins. During these game states, Ryan_FC successfully adopted positioning based on the intended implemented logic rather than default behaviours. This suggests that the designed routines were being triggered during set pieces.

Only 5 goals were scored from set pieces in the recorded matches, including 2 from freekicks, 1 from a corner and 2 from kick-ins. However, due to the small number of set piece occurrences, it doesn't provide a large enough sample to support detailed analysis

Also, while a structured routine was observed during set pieces, the final outcome of these routines depends on the subsequent decision making of individual agents. This means the effectiveness of set pieces can't be evaluated based only on its initial structure, due to the next phase often being open play where outcomes are created by agent interactions.

Overall, while direct goal output was limited by use case 2, the monitored evidence shows that the routines were successfully implemented at a structural level and contributed to improvements in organisation and possession retention.

Match Statistics and Visualisation

The custom analysis script (UC10) produced a set of visual outputs and statistics, including:

- Ball, team and player heatmaps
- Average formations
- Possession graphs
- Team level statistics (total goals scored, average goals scored etc)
- Player level statistics (total passes, average distance travelled etc)

The outputs supported the evaluation as they enabled direct comparison between Ryan_FC and HELIOS Base.

Qualitative Summary

The qualitative evaluation shows that the implemented tactics were both visible and effective within the RoboCup 2D Simulation environment. Across the recorded matches Ryan_FC constantly displayed structural and behavioural differences compared to the HELIOS Base team.

The findings showed that most of the use cases were added successfully including the defensive tactics (UC1,5,6,7), offensive tactics (UC8,9) and goalkeeper behaviour (UC3,4). These tactical behaviours weren't only observed, but can also be attributed to the improved performance displayed by Ryan_FC. Set Piece routines (UC2) were structurally implemented, however their overall effectiveness was difficult to quantify due to the limited number of occurrences and the reliance on agent decision making after moving into the correct structure. The stats and visualisations (UC10) helped to evaluate the successful implementation of the tactics and the effectiveness of them.

Overall, the qualitative results support the conclusion that Ryan_FC's tactical design was successfully created and contributed to the overall performance improvements seen in both the 15 game recorded matches as well as the 150 game quantitative analysis.

6.4 Discussion

The results of the quantitative and qualitative evaluation illustrate that Ryan_FC clearly improved in comparison to the baseline HELIOS Base team. The improvement was visible in both match outcomes

and in the structure and behaviour of the team. This indicated that the tactics implemented worked and massively impacted the performance.

The quantitative results show that Ryan_FC consistently outperformed HELIOS Base across all key areas, including win rate, goals scored, goals conceded, and possession. This suggests that the improvement was not due to isolated events, but rather sustained control of matches. The qualitative findings support this, illustrating that Ryan_FC maintained greater territorial dominance, applied sustained pressure in advanced areas, whilst remaining defensively stable.

A key finding was that the improvements came from a combination of tactics, including pressing (UC1), rest defence (UC7), 4-2-3-1 formation (UC5), and the double pivot (UC6), which all worked collaboratively to help control space more effectively. This balance between attacking pressure and defensive stability, facilitated higher team scores whilst simultaneously reducing goals conceded.

Spatial control was another major reason for improved attacking performance. Ryan_FC consistently spent more time in the opponent's half, final third, and penalty area, which in turn led to more high value scoring opportunities. This implies that occupying dangerous areas more often directly increases attacking chances.

The attacking behaviours (UC8 and UC9) provide clear evidence of successful tactical implementation. The team regularly moved the ball through wide areas before delivering it into central zones, reflecting the intended cutback based attacking pattern. The central attacking midfielder also effectively linked buildup plays to final execution. This indicates that these behaviours were both structured and repeatable rather than random.

However, not all implemented behaviours were equally effective. While the predictive goalkeeper and sweeper keeper behaviours (UC3 and UC4) were successfully implemented, the sweeper keeper was rarely used during buildup play. Similarly, set piece routines (UC2) were structurally present but had limited impact due to their low occurrence rate and reliance on subsequent agent decisions. These findings outline that implementing a tactic alone does not guarantee effectiveness.

These results show that real world football tactics can be implemented in the RoboCup 2D simulation environment. Tactics such as pressing, rest defence and cutback attacking patterns were not only added but also improved performance. While the simulation is not as complex as real football, it still proves that real world ideas can work well in this system.

Overall, the results show that Ryan_FC achieved its goal of being more effective than the HELIOS Base team. The improvements are supported by both stats and observed behavioural changes, clearly linking the tactics used to better performance.

6.5 Limitations and Future Work

While the evaluation shows that the implemented tactical behaviours improved performance in the RoboCup 2D simulation, there are several limitations to consider.

Firstly, the RoboCup simulation simplifies real football. It doesn't include physical differences between players such as speed and strength. It also doesn't include human decision making. This means all agents operate under the same physical limitations and are unable to think, adapt and improve like real players. As a result, the tactics may perform well in the simulation as it is predictable and controlled, rather than because they would be as effective in real football.

Another limitation is the reliance on rule based decision making within the system. Agents perform actions based on predefined conditions, instead of adapting to the match context. This can limit the effectiveness of certain tactics. For example, although the sweeper keeper behaviour was implemented, it was barely used in buildup play because the system prioritised forward passes over safer backward options. This shows that individual behaviours depend on the overall decision making framework.

The interaction between behaviours also presents a limitation. While multiple use cases were implemented, they don't always work together effectively. For example, in set pieces, a short pass may have happened as intended but the receiver chose to dribble or pass instead of shooting. This shows that even when a behaviour is triggered correctly, the lack of coordination between agents reduces the effectiveness of the behaviour.

There are also some limitations in how performance was measured. The statistics were created from custom analysis and do not directly measure specific behaviours. Stats such as possession and territorial control show that the team performed well, however they don't show which specific tactics caused the improvements. Some behaviours including cutbacks were identified by analysing player positions and movement which introduces a degree of subjectivity. Additionally, the RoboCup 2D log (rcg) files do not record all actions, meaning that some stats are estimates rather than exact measurements.

The qualitative analysis is limited by the sample size of recorded matches. Although 150 matches provided strong statistical reliability, behavioural analysis was based on only 15 games. This limits the ability to fully assess less frequent match instances, such as set pieces.

Future work could address these limitations by improving both agent decision making and evaluation methods. Integrating adaptive or learning based systems could allow agents to make context aware decisions, which would ensure behaviours such as sweeper keeper involvement and set piece routines are used more effectively during matches.

Further development of analysis tools could enable more precise and objective measurement of tactical behaviours, reducing reliance on estimation and interpretation, for example by more accurately

tracking passes and player actions. Increasing the number of monitored matches would improve the robustness of qualitative results, while introducing greater variability into the simulation, including player attributes such as speed and strength would provide a more realistic testing environment.

6.6 Broader Implications and Applications

This project shows that real world football tactics can be effectively translated into a simulated, agent based environment. The successful implementation of behaviours such as pressing, formations and cutback attacking patterns suggests that simulations can be used as a tool for analysing and testing tactical systems in a controlled setting.

This project has shown that it could be used for real world applications, for example in sports analytics and coaching. Simulation based systems could allow coaches and analysts to explore tactical ideas without the constraints and risks of real world matches, such as injuries. This could improve decision making and encourage a more data driven approach to tactical development.

There is also potential to turn this into an application for sports teams, such as analysis tools. These systems could help teams automatically analyse tactics, study opponents and support training sessions. However, for this to be successful, the simulation would need to be more realistic, including the addition of different players abilities, physical performance and more complex decision making.

While the results of the system show that tactical behaviours can improve performance within the system, it is important to recognise that the environment is simplified. Aspects of real football such as player physicality differences and technical ability, as well as human decision making aren't included in the simulation environment. This means that the results achieved should be seen as a close representation of real football rather than an exact match.

Finally, this work supports RoboCup's long term goal of developing intelligent multi-agent systems capable of effective coordination. Although the implementation is within the RoboCup 2D simulation, it demonstrates how structured tactical behaviours can improve coordination and overall performance.

7.0 Appendices

Appendix A – Project Repository

The full implementation of the system is available at:

[RyanMcL2004/Ryan-s-Final-Year-Project-Robocup](https://github.com/RyanMcL2004/Ryan-s-Final-Year-Project-Robocup)

This repository contains:

- Modified HELIOS Base codebase
- Implemented tactical behaviours
- Match analysis scripts and evaluation tools

Appendix B – Interactive Analysis Dashboard

Interactive analysis outputs are available at:

[Ryan_FC Analysis Dashboard](#)

This dashboard provides access to:

- Match report summaries (15 and 150 game datasets)
- Comparative analysis (Ryan_FC vs HELIOS Base)
- Heatmaps and positional data
- Full match recordings used for qualitative evaluation

Appendix C – Use of Generative AI

Generative AI tools (ChatGPT) were used in a limited capacity to support the preparation of this dissertation. This included assistance with improving clarity of written explanations, refining academic phrasing, and formatting references in IEEE style.

References

- [1] R. McLoughlin, “Ryan’s Final Year Project RoboCup,” GitHub repository. [Online]. Available: <https://github.com/RyanMcL2004/Ryan-s-Final-Year-Project-Robocup>. [Accessed: 27-Apr-2026].
- [2] P. Stone, “Robocup history: Robot soccer past and future,” IEEE Spectrum, Jun. 29, 2023. [Online]. Available: <https://spectrum.ieee.org/robocup-history>. [Accessed: 19-Dec-2025].
- [3] RoboCup Federation, “Media – FAQ,” RoboCup.org. [Online]. Available: https://www.robocup.org/media_faq. [Accessed: 24-Feb-2026].
- [4] I. Noda, H. Matsubara, K. Hiraki, and I. Frank, “Soccer server: A tool for research on multiagent systems,” Applied Artificial Intelligence, vol. 12, no. 2–3, pp. 233–250, 1998.
- [5] RoboCup Soccer Simulation League, “RoboCup Soccer Simulation League Home,” [Online]. Available: <https://ssim.robocup.org/>. [Accessed: 19-Dec-2025].
- [6] RoboCup Soccer Simulator Developers, “The RoboCup Soccer Simulator Users Manual,” [Online]. Available: <https://rcsoccersim.readthedocs.io/>. [Accessed: 25-Feb-2026].
- [7] D. M. Budden, P. Wang, O. Obst, and M. Prokopenko, “RoboCup simulation leagues: Enabling replicable and robust investigation of complex robotic systems,” arXiv:1412.5711, 2014. [Online]. Available: <https://doi.org/10.48550/arXiv.1412.5711>.
- [8] H. Akiyama, T. Nakashima, K. Hatakeyama, T. Fujiwara, and A. Hishiki, “HELIOS2024: Team Description Paper,” RoboCup Soccer Simulation League, 2024. [Online]. Available: https://ssim.robocup.org/wp-content/uploads/2024/07/TDP_HELIOS2024.pdf. [Accessed: 19-Dec-2025].
- [9] H. Akiyama, H. Shimora, and I. Noda, “HELIOS2008 Team Description,” Information Technology Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Ibaraki, Japan, 2008.
- [10] RoboCup Soccer Simulation League, “2D Awards,” RoboCup Soccer Simulation League. [Online]. Available: <https://ssim.robocup.org/soccer-simulation-2d/2d-awards/>. [Accessed: 01-Mar-2026].
- [11] H. Akiyama and T. Nakashima, “HELIOS base: An open source package for the RoboCup soccer 2D simulation,” in *RoboCup 2013: Robot World Cup XVII*, S. Behnke et al., Eds. Berlin, Germany: Springer, 2014, pp. 528–535.
- [12] N. Zare, O. Amini, A. Sayareh, M. Sarvmaili, A. Firouzkouhi, S. R. Rad, A. Soares, and S. Matwin, “Cyrus2D base: Source code base for RoboCup 2D soccer simulation league,” arXiv preprint arXiv:2211.08585, 2022. doi: 10.48550/arXiv.2211.08585.
- [13] H. Kitano, M. Asada, Y. Kuniyoshi, I. Noda, and E. Osawa, “RoboCup: A challenge problem for artificial intelligence,” AI Magazine, vol. 18, no. 1, pp. 73–85, 1997.

- [14] RoboCup Federation, "Objective," RoboCup.org. [Online]. Available: <https://www.robocup.org/objective>. [Accessed: 01-Mar-2026].
- [15] F. R. Goes, M. S. Brink, M. T. Elferink-Gemser, M. Kempe, and K. A. P. M. Lemmink, "The tactics of successful attacks in professional association football: Large-scale spatiotemporal analysis of dynamic subgroups using position tracking data," *Journal of Sports Sciences*, 2020. doi: 10.1080/02640414.2020.1834689.
- [16] R. Marcelino, J. Sampaio, G. Amichay, B. Gonçalves, I. D. Couzin, and M. Nagy, "Collective movement analysis reveals coordination tactics of team players in football matches," *Chaos, Solitons & Fractals*, vol. 138, 2020. doi: 10.1016/j.chaos.2020.109831.
- [17] H. L. Chen, R. Beumer, E. Torta, H. Bruyninckx, and R. van de Molengraft, "Designing offensive robot soccer strategies from principles in invasion team sports," *Journal of Intelligent & Robotic Systems*, vol. 112, art. no. 2, 2026.
- [18] J. Young, "Implementation of real world modern football tactics in RoboCup soccer simulator," B.Sc. dissertation, School of Computer Science, Queen's University Belfast, Belfast, U.K., Apr. 2025.
- [19] Z. Li, "Comparative study on the characteristics of technical and tactical application in different football formations," *Journal of Medicine and Physical Education*, vol. 1, no. 5, pp. 47–53, 2024.
- [20] A. Clarke, "Season trends: What were the most-popular formations in 2024/25?" *Premier League*, Jun. 5, 2025. [Online]. Available: <https://www.premierleague.com/en/news/4323030>. [Accessed: 01-Mar-2026].
- [21] A. Tweedale, "The 4-2-3-1: football tactics explained," *Coaches' Voice*, Aug. 8, 2022. [Online]. Available: <https://learning.coachesvoice.com/cv/the-4-2-3-1-football-tactics-pochettino-guardiola-flick-southgate/>. [Accessed: 01-Mar-2026].